Tunnelling and Underground Space Technology 179 (2027) 108096



Contents lists available at ScienceDirect

Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust



Numerical investigation of repurposing oil storage caverns for hydrogen storage using lined rock cavern (LRC) technology☆

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ARTICLE INFO

Keywords:
Lined rock cavern (LRC)
Oil storage cavern
Uplift failure
Numerical investigation
Cost analysis

ABSTRACT

The growing demand for large-scale geological hydrogen storage has driven the exploration of resource-efficient methods for repurposing existing underground infrastructure. This study assesses the technical feasibility of converting underground oil storage caverns into lined rock caverns (LRCs) for hydrogen storage and evaluates the material costs associated with two proposed design schemes. Two retrofit schemes were analyzed numerically using parameters derived from an existing LRC demonstration project. The ultimate and maximum operating gas pressures were determined using the uplift failure model, considering cavern burial depth and the mechanical properties of the surrounding rock mass. Structural responses under pressurization were evaluated, followed by a comparative assessment of concrete and steel liner costs. The results indicate that dividing the original cavern into multiple lined sub-caverns separated by concrete pillars improves structural performance and reduces the maximum tensile stress in the steel liner, although tensile-stress concentrations may develop in the concrete pillars. Meanwhile, the proposed Single-cavern scheme offers significantly higher hydrogen storage capacity with lower concrete requirements but a larger steel liner area. A supplementary short-term cyclic thermo-mechanical analysis revealed response-dependent differences from the corresponding static analysis. The static analysis overestimated the maximum tensile stress in the steel liner and the plastic-zone volume in the concrete liner, while underestimating the plastic-zone volumes in the rock-mass and total system. These findings highlight the importance of cyclic thermo-mechanical effects on LRC response while supporting the applicability of static analysis as a preliminary assessment approach, provided that an adequate safety margin is adopted when determining the maximum operating gas pressure. Additionally, in both schemes, the tensile-stress level in the steel liner remains low, suggesting significant potential for liner optimization in material selection and liner thickness. Overall, the results indicate that repurposing oil storage caverns as LRCs is a technically feasible retrofit option, while overall economic feasibility should be further evaluated through life-cycle cost analysis.

1. Introduction

Against the backdrop of the global transition toward low-carbon energy systems, hydrogen has emerged as a potential secondary energy carrier because of its high gravimetric energy density and potential for low-carbon use (Kovač et al., 2021). With the increasing penetration of renewable energy sources, hydrogen is not only regarded as an effective medium for energy storage but also offers a promising solution to seasonal imbalances in energy supply and demand (Evro et al., 2024). Among the various hydrogen storage methods, underground hydrogen storage has attracted growing attention due to its combined advantages in capacity, cost-effectiveness, and safety (Zhang et al., 2024). In particular, lined rock caverns (LRCs), which are suitable for hard-rock settings, are a viable and flexible option for underground hydrogen storage in areas lacking salt caverns or depleted oil/gas fields, because their steel sealing liner and concrete support system provide adaptable gas containment in competent rock (Masoudi et al., 2024). LRCs rely primarily on the surrounding rock mass to withstand internal gas pressure and prevent structural failure, while the primary function of the liner system is to ensure gas-tight sealing. The lined rock cavern (LRC)

☆ This article is part of a special issue entitled: 'ACUUS 2025' published in Tunnelling and Underground Space Technology incorporating Trenchless Technology Research.

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https://doi.org/10.1016/j.tust.2026.108096
Received 19 January 2026; Received in revised form 26 August 2026; Accepted 31 August 2026
Available online 7 September 2026
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technology was first tested in 1988 within the Grängesberg pilot project in Sweden, where several small-scale vertical cylindrical caverns were constructed to investigate high-pressure gas storage for natural gas applications. Pressure tests reaching several tens of MPa under cyclic loading conditions demonstrated stable rock mass behavior and confirmed the mechanical feasibility of the LRC concept (Johansson, 2003; Mansson et al., 2006). Subsequently, a full-scale lined rock cavern for pressurized natural gas storage was developed and tested in the Skallen project in Sweden. The cavern is a vertical cylinder with a domed top and rounded base, measuring 36 m in diameter and 52 m in height, and situated at a depth of approximately 115 m. Operation at pressures up to approximately 22 MPa verified the technical viability of LRCs at commercial scale and provided a benchmark case for numerical modeling and design validation (Glamheden and Curtis, 2006; Mansson et al., 2006). More recently, between 2022 and 2025, the technology was further demonstrated in a pilot-scale lined rock cavern with a volume of approximately 100 m^3 in Luleå, Sweden, where the results indicated promising performance for hydrogen storage (Vattenfall Press Office, 2025). New LRCs are typically designed for favorable mechanical performance using a vertical cylindrical form with a domed roof and a rounded or domed floor. Typical dimensions range from approximately 60–100 m in height and 30–40 m in diameter. To prevent uplift failure, LRCs are generally located at depths of about 100–200 m below ground surface, corresponding to roughly three times the cavern diameter (Johansson, 2003; Johansson et al., 2018; Mansson et al., 2006; Semprich et al., 1990; Sofregaz and Gustafsväg, 1999).

However, the construction of new LRCs is associated with substantial costs (Papadias and Ahluwalia, 2021). Accordingly, increasing research attention has focused on repurposing unused or abandoned underground spaces as lined energy storage caverns to avoid new cavern excavation and reduce construction costs (Liu et al., 2024; Saigustia and Robak, 2021; Schmidt et al., 2020; Wu et al., 2026). Because of their large storage volumes and existing excavation infrastructure, these spaces are regarded as promising candidates for compressed air and hydrogen storage. Within the framework of the project *Hydrogen, Ammonia, and Methanol in Hydrogen Hubs in the Nordic Region* (H2AMN), the feasibility of converting decommissioned oil storage caverns into Lined Rock Caverns (LRCs) for the storage of compressed hydrogen gas across different types of Nordic ports is being investigated (Lidberg et al., 2024). In Scandinavia, underground oil-storage caverns are typically constructed as shallow, unlined rock caverns with a horseshoe-shaped cross-section, commonly located at depths of less than 100 m beneath the ground surface and having typical dimensions of about 15–30 m in width, 25–40 m in height, and 100–500 m in length (Nilsen, 2021). These oil-storage caverns differ markedly in geometry and dimensions from newly constructed lined rock caverns, particularly in that their length is significantly greater than their width, which poses substantial challenges for LRC retrofitting. Moreover, the relatively shallow depth of such caverns at port locations provides limited confinement from the surrounding rock mass, thereby constraining the maximum operating gas pressure and potentially compromising overall cavern stability under high internal pressures.

When assessing the stability of lined rock caverns, a key concern is the prevention of uplift failure of the overlying rock mass under high internal gas pressure. Numerous studies have suggested that uplift failure of the rock mass occurs in the form of a truncated cone and the ultimate gas pressure under extreme conditions has been assessed using an uplift failure model. The maximum operating gas pressure is lower than the ultimate gas pressure because engineering projects usually require a certain safety margin. Littlejohn and Bruce (1975) developed a rigid-cone limit-equilibrium model to estimate uplift resistance, proposing cone angles of 60° or 90°. This approach was later applied by Brandshaug et al. (2001) to evaluate the safety of the gas storage cavern at the Skallen demonstration plant in Sweden. Subsequently, several studies have refined this model to improve estimates of the ultimate gas pressure (Damasceno et al., 2020; Kim et al., 2012; Pan and Tang, 2025; Perazzelli and Anagnostou, 2018; Tunsakul et al., 2018; Wang et al., 2022, 2023). Although the predicted results vary, these studies consistently indicate that uplift resistance is primarily governed by rock mass strength, cavern burial depth, in-situ stress as well as cavern geometry and layout (horizontal versus vertical). Furthermore, they conclude that while LRCs may be constructed at relatively shallow depths (e.g., 30–50 m), the maximum operating gas pressure must be correspondingly reduced (Perazzelli and Anagnostou, 2016).

In addition to uplift failure assessment, the overall stability of an LRC system also depends on the mechanical performance of the steel liner, concrete liner, and surrounding rock mass under operational gas pressure. In LRC systems, the surrounding rock mass provides the primary load-bearing capacity against high internal gas pressure, while the steel liner mainly serves as the gas-sealing barrier. Therefore, the maximum operating gas pressure must be maintained below the system's structural bearing capacity to ensure sufficient safety margins against rock instability, excessive deformation, and potential structural damage (Stille et al., 1994; Kim et al., 2012; Yi et al., 2024). Cracking of the rock mass or the concrete liner, as well as localized stress or strain concentrations in the steel liner may compromise the long-term stability and gas-tight integrity of the storage system. Previous studies on underground hydrogen storage have mainly focused on deeper geological formations, whereas the feasibility and mechanical performance of shallow caverns converted from existing oil storage tunnels remain insufficiently understood. To address this gap, this study investigates the feasibility of converting shallow oil storage caverns under Nordic conditions into LRCs for hydrogen storage.

This study proposes two retrofit schemes for converting shallow underground oil storage caverns into hydrogen storage facilities and subsequently evaluates their feasibility through analytical and numerical approaches. The primary analyses in this study focus on static mechanical stability and a preliminary material-cost assessment. As a supplementary investigation, Section 6.1 of the Discussion presents a short-term cyclic thermo-mechanical analysis. Transient gas-pressure and gas-temperature histories calculated using a thermodynamic model are applied to the mechanical model to compare the cyclic and static responses (Rutqvist et al., 2012; Zhou et al., 2015, 2017; Jiang et al., 2020; Geng et al., 2024). Two representative design schemes are considered: (i) the Single-cavern scheme, in which one continuous LRC is constructed along the cavern length, and (ii) the Multi-cavern scheme, in which multiple LRCs are separated by concrete pillars. Existing uplift failure models are first reviewed and compared, after which the selected model is extended to estimate the ultimate gas pressure of shallow elongated caverns. The corresponding maximum operating gas pressures are then determined using appropriate engineering safety factors. Based on the determined operating pressures, three-dimensional numerical models are established to investigate the mechanical responses of the steel liner, concrete liner, and surrounding rock mass. A parametric sensitivity analysis is then conducted to evaluate the influence of key engineering and geological factors, including rock mass quality, cavern depth, in-situ stress ratio, steel liner thickness, spacing between adjacent sub-caverns, and cavern length. Finally, the proposed retrofit schemes are compared in terms of structural response, hydrogen storage capacity, and preliminary material cost, providing a quantitative basis for assessing the technical feasibility of repurposing shallow oil storage caverns for underground hydrogen storage.

2. Retrofitting schemes and uplift failure analysis

2.1. Proposed retrofitting schemes

The majority of existing underground oil storage caverns in Sweden are situated at depths of approximately 20–40 m, close to ports, and have lengths exceeding 100 m. According to the schematic diagram of the typical cavern layouts proposed by Jiang et al. (2024), two modification schemes are proposed in this study, as illustrated in Fig. 1. The

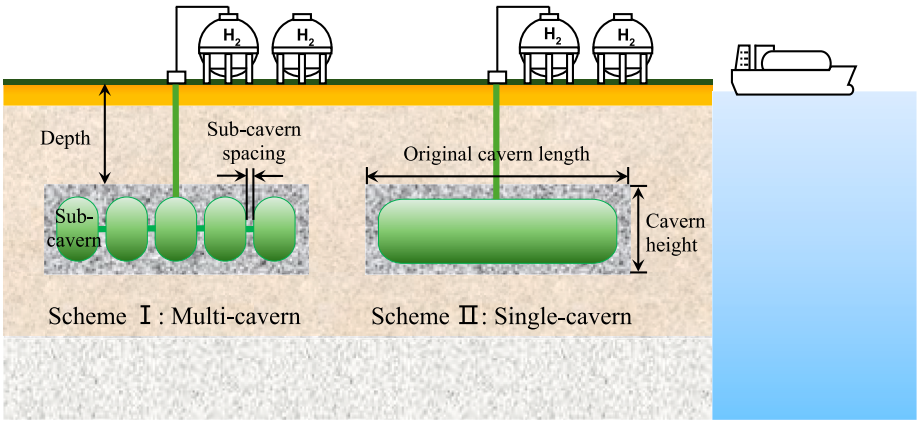


Fig. 1. Schematic diagram of two schemes for converting underground oil storage caverns into hydrogen storage LRCs.

first is a Multi-cavern scheme (Scheme I), in which the original unlined cavern is divided into several compartments by concrete pillars, and a vertical cylindrical LRC with a hemispherical dome and floor is constructed for each compartment. A hemispherical floor geometry is adopted in this study, as it provides superior mechanical performance following optimization by Luo et al. (2024). The second is a long Single-cavern scheme (Scheme II), in which a continuous elongated LRC is constructed along the entire length of the unlined cavern. In the Single-cavern scheme, the end sections of the LRC are shaped as half-cylindrical segments. To avoid ambiguity, the term "cavern" refers to an unlined rock cavern with exposed rock surfaces, whereas the terms Multi-cavern, sub-cavern, and Single-cavern denote LRCs with concrete and steel liners. A lined cavern is slightly smaller in size than the original cavern.

2.2. *Uplift failure analysis of pressurized rock caverns*

For shallow pressurized rock caverns, uplift failure of the overlying rock mass is one of the primary constraints controlling the ultimate gas pressure. When the internal gas pressure exceeds the combined resistance provided by the self-weight, strength, and confinement of the overburden, the rock mass above the cavern may be progressively fractured and lifted, leading to a loss of structural stability. This issue is particularly critical for the retrofitting schemes considered in this study, because existing oil storage caverns in Sweden are generally shallow and elongated, and therefore provide limited overburden confinement compared with newly developed LRCs that are constructed at depths sufficient to prevent uplift failure. Accordingly, uplift failure analysis is first conducted to estimate the ultimate gas pressure before evaluating the mechanical response of the liner–concrete–rock system under operational pressure.

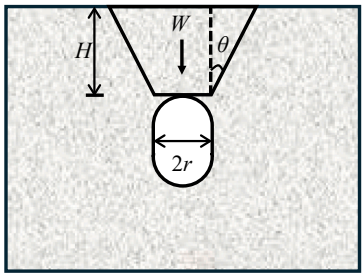
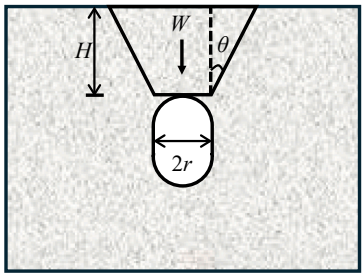
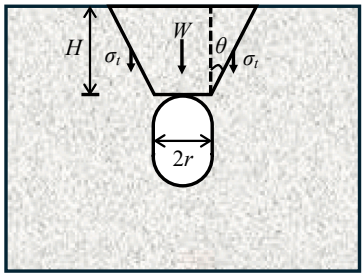
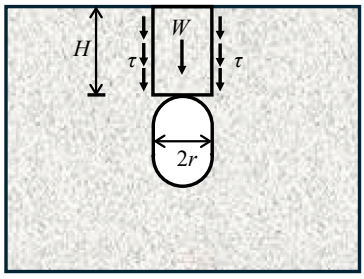
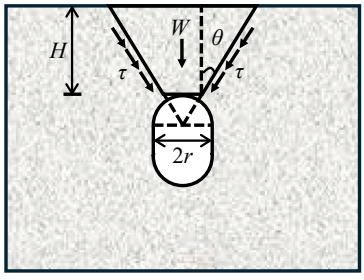
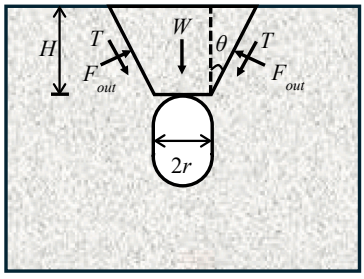
2.2.1. *Comparison of existing uplift failure models*

As research on pressurized storage caverns has progressed, numerous analytical models have been proposed to evaluate uplift failure. Among these, two limit equilibrium models, the rigid-cone model and the log-spiral model, have been widely used to determine the ultimate gas pressure or the minimum cavern burial depth required to prevent overlying rock mass failure (Brandshaug et al., 2001). Six uplift failure analysis models are summarized and listed in Table 1. The six uplift failure models are evaluated in a uniform two-dimensional form. Moreover, the Multi-cavern and Single-cavern schemes are assumed to exhibit equivalent uplift mechanisms. This study focuses on the influence of cavern burial depth and rock mass quality on the ultimate gas pressure predicted by these models, enabling comparison of model predictions and supporting model selection.

The rigid-cone model assumes a straight, inclined upward failure plane extending to the ground surface, with the overlying rock mass idealized as a conical body. According to Littlejohn and Bruce (1975), the cone angle is typically taken as either 90° or 60°, with the latter mainly applied to soft, heavily fissured, or weathered rock masses. In this model, the uplift resistance is equated solely to the self-weight of the rock cone; consequently, the strength of the rock mass is not explicitly considered. The log-spiral model originates from studies on the pull-out resistance of shallow soil anchors (Ghaly and Hanna, 1994). It assumes that uplift resistance is mobilized through frictional sliding along a curved logarithmic spiral failure surface. However, the development of such a mechanism requires sufficient lateral confinement to maintain shear resistance along the failure path. Mandl (1988) demonstrated that decreasing horizontal stress results in narrower and steeper spiral geometries, indicating a transition toward more localized uplift failure. Consequently, under low in-situ horizontal stress conditions, the conventional log-spiral criterion may overestimate uplift resistance because the assumed frictional failure mechanism cannot be fully mobilized (Brandshaug et al., 2001). This limitation is particularly relevant for shallow elongated caverns, where overburden confinement is relatively limited and uplift failure may be governed by localized tensile-shear failure rather than by a fully developed rotational spiral mechanism.

Given these considerations, this study adopts a straight, upward failure plane extending to the ground surface, consistent with assumptions used in previous research. Building upon the rigid-cone model, Murray and Geddes (1987) proposed replacing the geometric cone angle with the internal friction angle of the rock mass. Vezole (2002) considered the tensile cutoff in the uplift direction (vertical direction) in his solution for soil materials to resist uplift forces. Based on these developments, Damasceno et al. (2020) modified the rigid-cone model to evaluate the ultimate gas pressure in LRCs. Moreover, Kim et al. (2012) proposed an analytical model in which the failure surface initiates at the cavern crown and propagates vertically upward toward the ground surface, accounting for both the self-weight of the overlying rock mass and the shear resistance along the failure plane. Based on experimental, analytical, and numerical investigations, Tunsakul et al. (2013), Tunsakul et al. (2014), 2018) found that the initiation point of cavern roof uplift failure is strongly influenced by the in-situ stress ratio. Accordingly, they proposed an uplift failure model by assuming a failure plane originating from the failure initiation point and combining the rigid-cone concept with the shear resistance developed along the failure surface. Xia et al. (2023) and Pan and Tang (2025) further investigated crack initiation and propagation around pressurized CAES caverns using thermo-mechanical phase-field modeling and upper-bound limit analysis, respectively. Their studies indicate that fracture initiation locations and propagation paths depend on the in-situ stress state and rock mass properties rather than following a single predefined geometry. In addition, Wang et al. (2023) developed a model in which the failure surface is assumed to propagate along a straight and inclined plane extending toward the ground surface, explicitly considering the associated shear resistance along the failure interface.

Table 1
Existing uplift failure models in 2D form.

Uplift failure model	Description	Reference
	Model 1: The failure surface is assumed to be a straight, inclined upward failure plane extending to the ground surface. The half-cone angle (α) between the failure surface and the vertical direction is typically taken as either 45° or 30°, with the latter mainly applied to soft, heavily fissured, or weathered rock masses. Here, $\theta = \alpha$. $P = \gamma H(r + 0.5H\tan\alpha)/r$	Modified from Littlejohn and Bruce (1975)
	Model 2: A straight failure surface is assumed, inclined to the vertical by an angle equal to the internal friction angle of the rock mass: $\theta = \varphi$. $P = \gamma H(r + 0.5H\tan\varphi)/r$	Modified from Murray and Geddes (1987)
	Model 3: The angle between the failure surface and the vertical direction is assumed to equal the internal friction angle of the rock mass: $\theta = \varphi$. The resistance effect of the rock mass tensile strength (σ_t) is also considered. $P = [\gamma H(r + 0.5H\tan\varphi) + \sigma_t(H\tan\varphi)]/r$	Modified from Damasceno et al. (2020)
	Model 4: The potential failure surface is oriented vertically upwards: $\theta = 0°$ and the shear resistance (τ) of the failure surface is considered. c denotes the cohesion of the rock mass. $P = \gamma H + (0.5\lambda\gamma H^2\tan\varphi + cH)/r$	Modified from Kim et al. (2012)
	Model 5: The failure surface extends from the initial crack location to the ground surface, $\theta = \beta$. β is the crack initiation angle. $P = \frac{\gamma H(2R + H)}{2r} + \frac{cH/\cos\beta + \gamma H^2\lambda\tan\varphi}{r\tan\beta}$ $\beta = 31.50\ln(\lambda) + 38.58(^\circ)$	Modified from Tunsakul et al. (2018) and Pan and Tang (2025)
	Model 6: The angle between the failure surface and the vertical direction is set as $\theta = 45° - \varphi/2$. F_{out} is the total pressure on the failure surface, and T is the total shear force on the failure surface. $P = (W + 2T\cos\theta - 2F_{out}\sin\theta)/2r$ $W = \gamma[2r + H\tan(45° - \frac{\varphi}{2})]H$ $T = F_{out}\tan\varphi + cH\sec\theta$ $F_{out} = \frac{0.5\lambda\gamma H^2 - cH\tan\theta}{\cos\theta + \tan\varphi\cdot\sin\theta}$	Wang et al. (2023)

(continued on next page)

Table 1 (*continued*)

Uplift failure model	Description	Reference

Note: P is the ultimate gas pressure, MPa; γ is the unit weight of rock mass, MN/m^3; H is the cavern burial depth, m; r is half of the width (or radius) of the cavern, m; θ is the failure angle between the potential failure surface and the vertical direction, °; λ is the in-situ stress ratio.

Although straight failure geometries generally yield smaller overlying rock mass weights than curved failure planes (e.g., Wang et al., 2022; Pan and Tang, 2025), they offer a practical advantage of avoiding uncertainties related to the shape of curved failure surfaces, which are often difficult to determine reliably in advance.

Based on the description of the oil storage cavern in Section 2.1, the width of the shallow buried long cavern is taken as $2r = 18$ m and the in-situ stress ratio λ is assumed to be 1. A detailed model description is shown in Fig. 4. The rock mass parameters refer to the Skallen demonstration project in Sweden. The intact rock properties include a Young's modulus (E_i = 57 GPa), a material constant (m_i = 25), a disturbance factor (D = 0), Poisson's ratio (ν = 0.25) and a uniaxial compressive strength (UCS = 150 MPa) (Damasceno et al., 2023; Glamheden and Curtis, 2006; Hoek et al., 2002; Hoek and Diederichs, 2006). In this study, the method for calculating cohesion and friction angle was adopted according to Zhang et al. (2019), whereas Young's modulus and tensile strength are estimated based on Hoek et al. (2002). Since the six models in Table 1 consider the influence of the in-situ stress ratio differently, $\lambda = 1$ was assumed here to ensure consistency in the comparison. With a cavern burial depth of 30 m, the parameters of the Mohr-Coulomb model under different Geological Strength Index (*GSI*) conditions are calculated and shown in Table 2.

Fig. 2a illustrates the variation of the ultimate gas pressure with depth as predicted by different uplift failure models. When burial depth was varied, *GSI* was fixed at 65; when *GSI* was varied, burial depth H was fixed at 30 m. In all models, the ultimate gas pressure increases monotonically with depth, highlighting the dominant role of depth in enhancing the cavern confinement. However, significant differences exist among the models. Model 1 and Model 2 predict relatively lower pressure levels. In contrast, Models 4, 5, and 6 consistently yield higher ultimate gas pressures across the entire depth range, mainly attributable to the consideration of cohesion of the rock mass. Model 3 predicts intermediate pressure levels. As depth increases, the differences in the predicted pressures among the models become more pronounced, which would lead to substantial variations in the design of LRCs. Fig. 2b shows the effect of rock mass quality, expressed by the *GSI*, on the ultimate gas pressure. Except for Model 1 and Model 2, all models indicate a clear positive correlation between ultimate gas pressure and *GSI*, confirming that the load-bearing capacity of the cavern increases significantly with improving rock mass quality. Similarly, Model 3 predicts intermediate ultimate gas pressures among all the models. With H = 30 m and GSI = 65 as baseline parameters, Model 3 predicts an ultimate gas pressure that is 267 % and 176 % of those predicted by Models 1 and 2, respectively, and 41 %, 23 %, and 49 % of those predicted by Models 4, 5, and 6, respectively.

Given the shallow burial depth of the LRC, Model 3 was adopted as a screening model for estimating the ultimate gas pressure. It produces intermediate estimates among the six models and accounts for both the self-weight and tensile strength of the overlying rock mass. Its predictions are therefore interpreted as comparatively conservative relative to the higher estimates from Models 4–6, rather than as a lower-bound solution among all available models.

Table 2
Mohr-Coulomb parameter values corresponding to the *GSI* values used in the analysis.

Geological Strength Index	Elastic modulus	Poisson's ratio	Cohesion	Friction angle	Tensile strength
GSI	E (GPa)	ν	c (MPa)	φ (°)	σ_t (MPa)
35	6.46	0.25	1.33	55.98	0.04
40	9.10	0.25	1.48	57.31	0.07
45	12.75	0.25	1.66	58.47	0.09
50	17.51	0.25	1.89	59.48	0.14
55	23.27	0.25	2.18	60.31	0.20
60	29.64	0.25	2.58	60.95	0.29
65	36.01	0.25	3.14	61.39	0.43
70	41.77	0.25	3.91	61.61	0.62
75	46.53	0.25	4.99	61.59	0.91
80	50.18	0.25	6.51	61.34	1.33
85	52.82	0.25	8.62	60.88	1.94
90	54.64	0.25	11.56	60.22	2.82
95	55.87	0.25	15.64	59.38	4.12

2.2.2. Three-dimensional theoretical ultimate gas pressure

The impact of cavern length on gas storage capacity and material cost, as analyzed in Section 5.2, warrants further consideration. Accordingly, Model 3 is further extended into a three-dimensional form. As illustrated in Fig. 3, a modified uplift failure formulation applicable to elongated caverns is derived based on this model.

The volume of the rock mass, V_r, is given by:

$$V_r = \frac{1}{3}H(A_1 + A_2 + \sqrt{A_1 A_2}) \tag{1}$$

where A_1 is the area of the top surface of the overlying rock mass, m^2; A_2 is the area of the bottom surface of the overlying rock mass, m^2.

The cross-sectional area in the tensile area, A_T, is given by:

$$A_T = A_1 - A_2 = (L_c + 2H\tan\varphi) \times (W_c + 2H\tan\varphi) - L_c \times W_c \tag{2}$$

where L_c and W_c are the cavern length and width, respectively.

On surface A_2, the uplift force induced by gas pressure equals the weight of the rock mass and its tensile resistance. Accordingly, for shallow elongated caverns, the ultimate gas pressure predicted by Model 3 under the ultimate limit state (F_{sf} = 1) can be calculated as follows (modified from Damasceno et al., 2020):

$$P = \frac{V_r\gamma + \sigma_t A_T}{A_2 F_{sf}} = \frac{\frac{1}{3}\gamma H(A_1 + A_2 + \sqrt{A_1 A_2})}{A_2 F_{sf}} + \frac{\sigma_t(A_1 - A_2)}{A_2 F_{sf}} \tag{3}$$

where F_{sf} is the safety factor. On the right-hand side of the equation, the first term represents the contribution of the weight of the overlying rock mass, and the second term accounts for the resistance provided by the tensile strength of the rock.

It should be noted that the valid application range of Eq. (3) is constrained by the assumptions and case conditions on which it was developed. In particular, its applicability is expected to depend on the tensile failure of the overlying rock mass and factors such as cavern geometry, layout, burial depth, rock mass quality, and in-situ stress conditions.

The modified uplift failure model provides a useful first-order estimate of the ultimate gas pressure for shallow, elongated caverns. However, it essentially represents an ultimate limit state and does not fully capture the progressive evolution of tensile failure in the overlying rock mass. Consequently, the predicted ultimate pressure should not be directly adopted as the operating pressure. An appropriate safety margin is required to account for uncertainties associated with rock-mass properties, failure mechanisms, and the predictive capability of the analytical model. As demonstrated by the comparison in Section 2.2.1, existing uplift models yield substantially different ultimate pressures under the same conditions, highlighting the uncertainty associated with

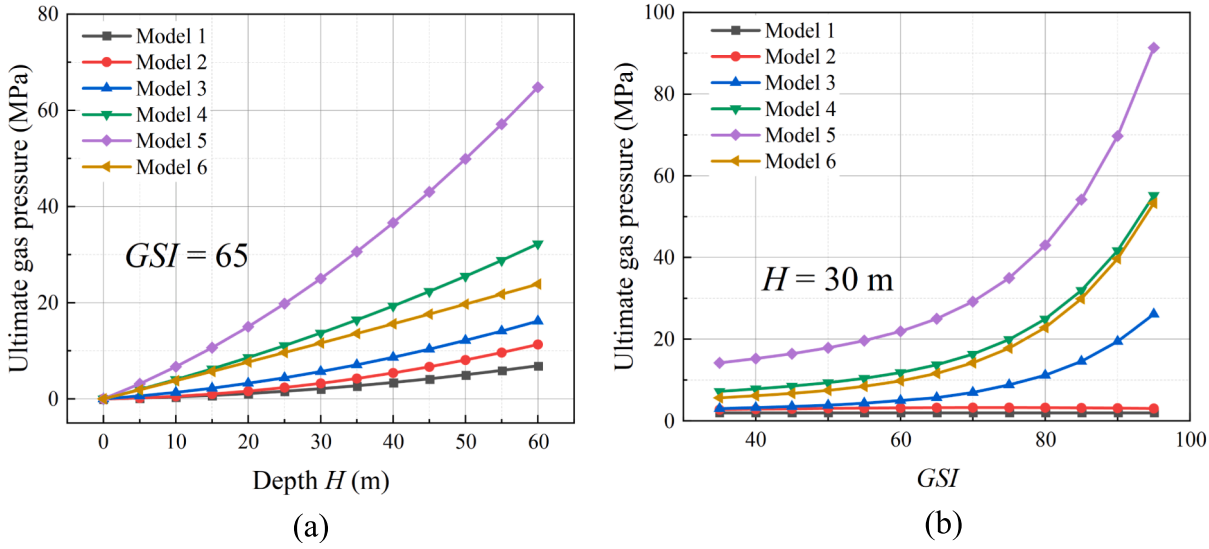


Fig. 2. Variation of ultimate gas pressure with (a) cavern burial depth H and (b) GSI.

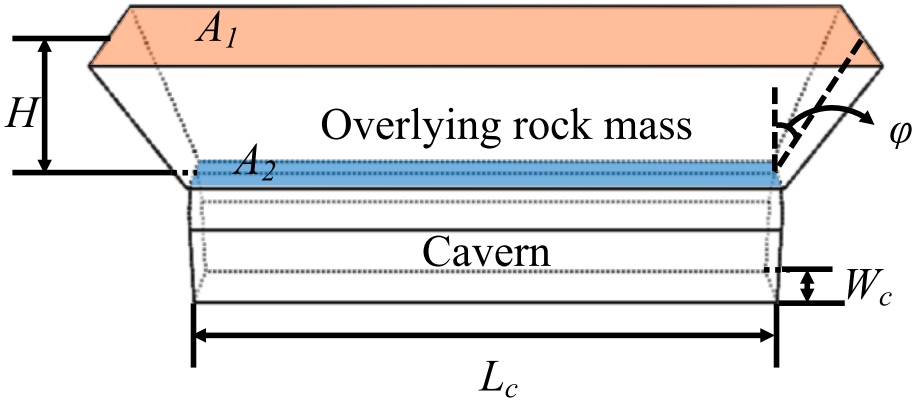


Fig. 3. Potential uplift failure in the overlying rock mass of a long cavern.

analytical prediction. Previous studies have similarly evaluated LRC uplift stability in terms of factors of safety that vary considerably with cavern geometry, burial depth, and rock-mass quality (Kim et al., 2012; Yi et al., 2024). Considering these uncertainties and the preliminary nature of the modified uplift model, a comparatively conservative value of $F_{sf} = 3$ was adopted in the present study to derive the maximum operating gas pressure from the predicted ultimate pressure. The supplementary cyclic thermo-mechanical analyses presented in Section 6.1 further reinforce the importance of maintaining an adequate safety margin, as the static analysis does not conservatively bound all response quantities.

It is worth noting that, owing to the limited number of shallow lined rock cavern projects, the predictive capability of existing uplift failure models still requires further verification under practical engineering conditions (Littlejohn and Bruce, 1975; Murray and Geddes, 1987; Kim et al., 2012; Tunsakul et al., 2018; Damasceno et al., 2020; Wang et al., 2022, 2023; Pan and Tang, 2025; Gan and Qin, 2026). Although several studies have attempted to examine these models through numerical simulations (Wang et al., 2022; Pan and Tang, 2025; Gan and Qin, 2026), the identification and assessment of uplift failure in numerical models remain challenging and are not yet standardized. Consequently, the robustness and reliability of current uplift failure models still need to be evaluated against field observations and monitored case histories.

At present, no published case is available for elongated lined rock caverns that could be used for direct validation of Eq. (3). However, a published cylindrical cavern case, namely the LRC demonstration project at Skallen, is available and was therefore used only as an order-of-magnitude reference for Model 3 under the corresponding geometry, rather than as a direct validation case. The cavern has a radius of 18 m and a burial depth of 115 m, with a maximum operating gas pressure of 20 MPa. The reported rock mass properties include a tensile strength of 0.7 MPa and a friction angle of 54.1° (Damasceno et al., 2023). Substituting these parameters into the three-dimensional cylindrical form of Model 3, Eq. (4) (Damasceno et al., 2020), gives a predicted ultimate gas pressure of 176 MPa, corresponding to a $F_{sf} = 8.8$ relative to the actual operating pressure:

$$P = \frac{\frac{\pi}{3}\gamma H\left[R^2 + (R + H\tan\varphi)^2 + R(R + H\tan\varphi)\right]}{\pi R^2} + \frac{\pi\sigma_t\left[(R + H\tan\varphi)^2 - R^2\right]}{\pi R^2} \tag{4}$$

This comparison indicates that Model 3, when directly applied to this deep cavern case, gives a high estimate. The large margin is likely related to the fact that the Skallen cavern lies at a substantially greater burial depth than the shallow conditions for which tensile uplift failure is of primary concern. Nevertheless, this published case shows that the predicted ultimate pressure exceeds the reported operating pressure in this example, but it does not establish predictive accuracy for shallow elongated caverns. Accordingly, Model 3 and its extended forms for

different geometries, such as Eqs. (3) and (4), may be used as a preliminary design tool, but their predictions should be interpreted together with engineering judgement and, where possible, supplemented by case-based calibration.

3. Numerical modeling

To evaluate the feasibility of converting shallow oil storage caverns into lined rock caverns for hydrogen storage, numerical models were established to simulate the mechanical response of the liner–concrete–rock system under maximum operating gas pressure based on retrofitting schemes. The model setup, including boundary conditions, constitutive relationships, and material parameters, was then described in detail to ensure the reliability and reproducibility of the analysis. Finally, a parametric sensitivity analysis was conducted to systematically investigate the influence of key geological and design factors on the structural performance of the system.

The numerical analyses presented in this section employ a static screening approach. The maximum operating gas pressure is applied as a constant internal load, while cyclic variations in cavern gas pressure and temperature are not considered. The implications of this simplification are further examined in Section 6.1 through comparison with supplementary short-term cyclic thermo-mechanical analyses.

3.1. Numerical model

Numerical simulations were performed using the explicit finite difference code FLAC3D 9.0 (Itasca Consulting Group, 2024). The numerical model was first constructed using Rhino 7.0 and the Griddle 2.0 plug-in and was then imported into FLAC3D. Based on the shallow unlined oil caverns studied in the H2AMN project and existing LRC designs, two schemes were simulated. The three-dimensional numerical model is shown in Fig. 4. Different element types were used to represent the main components of the LRC, including the rock mass, concrete liner, steel liner, and sliding layer. Specifically, solid elements were used to simulate the surrounding rock and concrete liner, while liner structural elements were used to simulate the steel liner. The sliding layer was represented by an interface between the concrete liner and the steel liner.

The two schemes differ mainly in whether the cavern system is divided by concrete pillars. In the Multi-cavern scheme, each sub-cavern features a hemispherical top and bottom with a radius of 8 m, and a vertical cylindrical middle section with a height of 7 m. The concrete between adjacent caverns, referred to as concrete pillars, had a thickness of 10 m and was also treated as the sub-cavern spacing in this study, as illustrated in Fig. 4a. In the Single-cavern scheme, the model is a single long cavern without concrete pillars along its length of 120 m, as shown in Fig. 4b. As shown in Fig. 4c, the thinnest part of the concrete in the cavern is the thickness of the concrete liner. These geometric definitions provide the basis for the subsequent mesh generation and loading conditions.

Following the definition of the model geometry, the numerical model was discretized using unstructured tetrahedral elements. A refined mesh with a uniform element size of 1 m was adopted in the region extending from the cavern boundary to the ground surface to accurately capture stress redistribution and plastic deformation in the overlying rock mass. In the remaining parts of the model, a coarser mesh with a maximum element size of 10 m was used to improve computational efficiency. Because the mesh strategy can influence the calculated plastic-zone volume, the same discretization approach was kept for all simulation cases. The same grid configuration was used in all cases to preserve comparability of the response trends. The adopted mesh density represents a compromise between computational efficiency and accuracy.

After the geometry and mesh were established, the excavation assumption, loading procedure, and boundary conditions were defined for numerical models. Because the existing oil storage cavern is pre-excavated, the study focuses on the performance of the converted hydrogen storage cavern and the plastic zone induced by excavation was not considered. In numerical models, the excavation process was simplified as a single-step excavation, followed by the installation of the liner system. Subsequently, the maximum operating gas pressure was applied directly to simulate the operating conditions of the converted

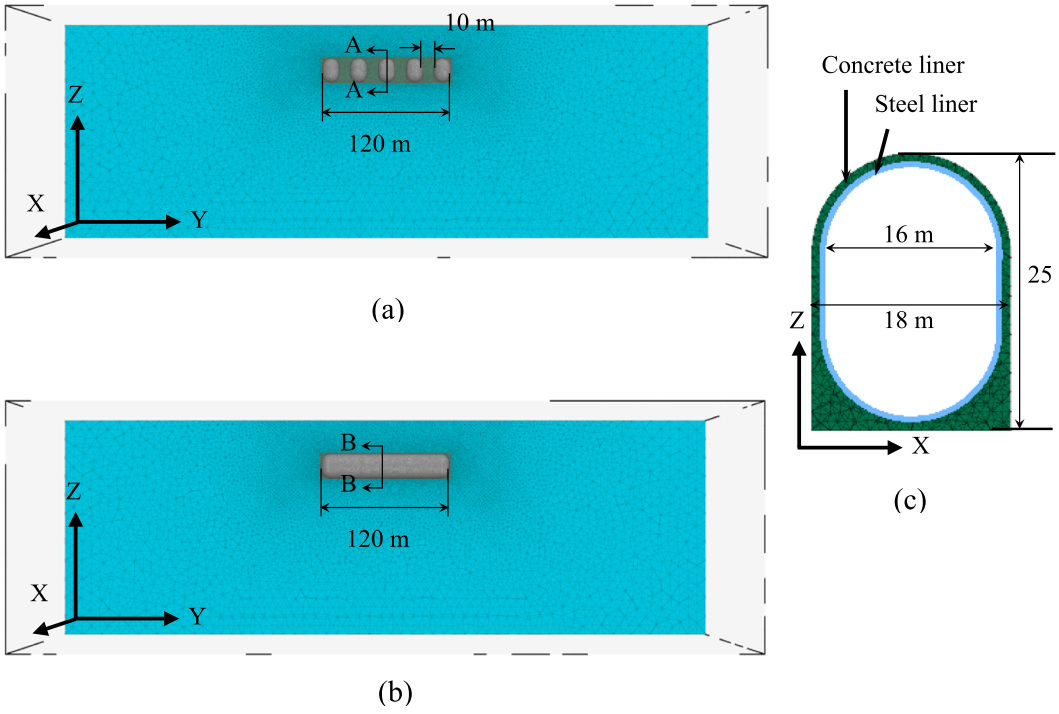


Fig. 4. Numerical models of lined rock caverns for two benchmark cases in two schemes: (a) Multi-cavern scheme; (b) Single-cavern scheme; and (c) cross-sections A–A and B–B.

lined rock cavern. The benchmark in-situ stress ratio λ was set to 1, and its influence was examined in the parametric sensitivity analysis. The top boundary of the model represents the unconstrained ground surface, while the lateral boundaries are modeled with roller constraints, and the bottom boundary is fully fixed. The major horizontal stress is oriented along the *Y*-axis of the numerical model, the minor horizontal stress along the *X*-axis, and the vertical stress along the *Z*-axis.

3.2. Input parameters

The Mohr–Coulomb model was adopted because its parameters have direct physical interpretations in both analytical and numerical assessments. The rock mass and concrete were simulated using elastic-perfectly plastic constitutive models, while the steel liner was modeled as an elastic material.

For the sensitivity analysis of rock strength, the *GSI* from the Hoek–Brown criterion was employed because it provides a direct representation of rock mass quality. The corresponding rock mass parameters are shown in Table 2.

The remaining parameters describe the liner system and the concrete-steel interface. The simulations included a C30 concrete liner and an S355 steel liner, with their properties listed in Table 3. The parameters for the concrete and steel liner were based on values reported in the Itasca report (Brandshaug et al., 2001). Considering the stress transfer effect at the interface between the steel liner and concrete due to potential slip, a friction coefficient of 0.1 was applied at the concrete-steel interface in the numerical models (Damasceno et al., 2023). The FLAC3D interface followed a Coulomb-sliding formulation (Itasca Consulting Group, 2024). Based on experimental and monitoring data reported by Johansson (2003), the parameters of the sliding layer, modeled as an interface, were determined through prior numerical simulations (Luo et al., 2024) and are listed in Table 3.

3.3. Parametric study design

The parametric study was organized into two parts so that the effects of Single-cavern design variables and the interaction of sub-caverns in the Multi-cavern scheme could be evaluated separately. First, a parametric sensitivity analysis was conducted based on the Single-cavern scheme to investigate the influence of key factors on system behavior. Previous studies on the evaluation of ultimate gas pressure using the uplift failure model have demonstrated that cavern depth, cavern radius, rock mass quality, and in-situ stress ratio are among the most influential parameters (Kim et al., 2012; Tunsakul et al., 2018; Pan and Tang, 2025). In this study, cavern depth, rock mass quality, and in-situ stress ratio were selected as the primary parameters for sensitivity investigation. In addition, the effects of steel liner thickness and concrete liner thickness were also considered to comprehensively assess the mechanical role of the liner system. Second, for the Multi-cavern scheme, the spacing between adjacent caverns was identified as a critical geometric parameter, and its influence on the interaction effect and overall stability was systematically examined. Table 4 lists the parameters used in the parametric study. One benchmark case was defined for each scheme. Both cases had $H = 30$ m, $d_c = 1$ m, $d_s = 0.012$ m, $GSI = 65$, and $\lambda = 1$. For the Multi-cavern benchmark case, the sub-cavern spacing was $S_c = 10$ m.

To ensure that the parametric comparisons reflect changes in the selected variables rather than changes in loading level, the same maximum operating gas pressure was used across all cases. This fixed-pressure setup was used only for the mechanical sensitivity analysis; in the cost analysis, pressure was recalculated from Eq. (3) for each H, GSI, and L_c. In the parametric study, $F_{sf} = 3$ was adopted as stated in Section 2.2.2. Therefore, according to Eq. (3), the maximum operating gas pressure of the benchmark case was calculated as 3.3 MPa.

Table 3
Parameters of the concrete liner, steel liner, and sliding layer (modified from Brandshaug et al. (2001) and Luo et al. (2024)).

	Parameters	Value	Unit
Concrete liner	Density	2400	kg/m^3
	Elastic modulus	30	GPa
	Poisson's ratio	0.2	–
	Cohesion	9.6	MPa
	Friction angle	45	°
	Tensile strength	2.7	MPa
Steel liner	Density	8000	kg/m^3
	Elastic modulus	200	GPa
	Poisson's ratio	0.3	–
Sliding layer	Normal stiffness	45,000	MPa/m
	Shear stiffness	43	MPa/m
	Friction coefficient	0.1	–

4. Results

The LRC relies on the steel liner to contain the compressed hydrogen gas. Under internal gas pressure, the steel liner is subjected to tensile stress. If the tensile stress reaches or exceeds the strength of the steel liner, liner failure may occur, potentially leading to hydrogen leakage. In addition, failure of the surrounding rock mass and the concrete liner may result in increased deformation and induce stress concentrations within the steel liner, thereby elevating the risk of failure. Moreover, damaged zones in the rock mass indicate the uplift risk of the overlying strata.

In the FLAC3D model, the maximum tensile stress in the steel liner can be monitored, and the extent of plastic zones can be quantified. Accordingly, the total plastic-zone volume and the corresponding volumes in the concrete liner and rock mass were calculated and compared for detailed analysis. Plastic-zone volumes were obtained by summing the volumes of all yielded zones in FLAC3D.

4.1. Comparison of two benchmark cases

A comparative analysis of the benchmark cases enables a preliminary assessment of the differences between the Multi-cavern and Single-cavern schemes. As shown in Fig. 5, the maximum tensile stress in the steel liner of the Multi-cavern scheme is 25.4 MPa, which is 78 % of the 32.4 MPa obtained for the Single-cavern scheme. In the LRC demonstration project in Skallen, Sweden, the design strain of the steel liner was 0.89 ‰ (Mansson et al., 2006). Based on the elastic modulus of the steel liner, the corresponding design stress is approximately 178 MPa. Both calculated stresses are well below this reference design stress. This comparison suggests scope for liner optimization, subject to dedicated checks of airtightness, fatigue, welding details, and material compatibility with hydrogen. In addition, the locations of maximum tensile stress differ between the two schemes. In the Multi-cavern scheme,

Table 4
Parameters used in parametric study.

Scheme	Factors	Symbol	Value	Unit
Multi-cavern	Sub-cavern spacing	S_c	2.5, 5, **10**	m
Single-cavern	Depth	H	20, **30**, 40	m
	Thickness of concrete liner	d_c	0.5, 0.75, **1**	m
	Thickness of steel liner	d_s	0.006, 0.009, **0.012**	m
	Geological strength index	GSI	50, **65**, 80	–
	In-situ stress ratio	λ	0.5, **1**, 2, 2.5, 3, 3.5	–

Note: Bold values denote the benchmark cases. In each sensitivity series, only one parameter was varied, while all other parameters were fixed at their benchmark values. Accordingly, combinations involving two or more non-benchmark parameter values were not considered. Different *GSI* values correspond to the rock-mass parameters listed in Table 2.

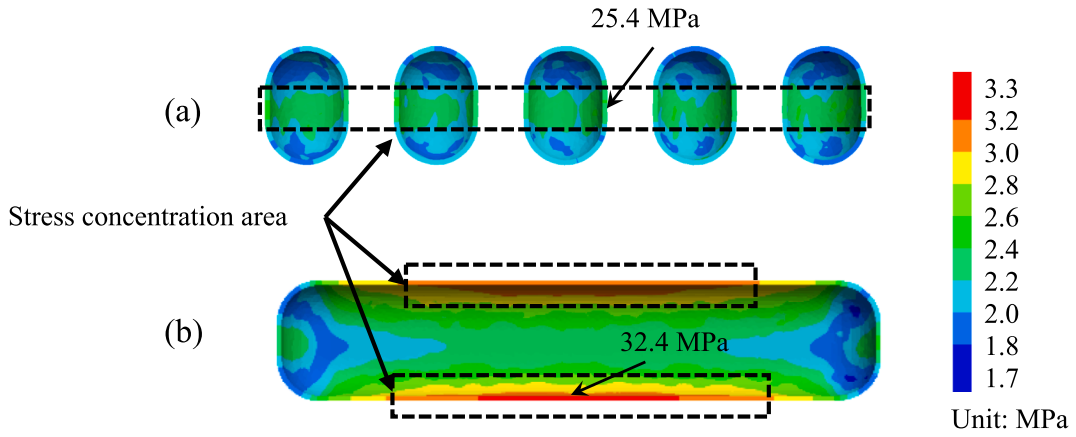


Fig. 5. Tensile stress distribution on the steel liner: (a) benchmark case of the Multi-cavern scheme; (b) benchmark case of the Single-cavern scheme.

tensile stress in the steel liner is primarily concentrated near the mid-height of the LRC and adjacent to the concrete pillars. In contrast, in the Single-cavern scheme, tensile stress concentrations occur at the top and bottom of the LRC, with comparable magnitudes at these locations.

As shown in Fig. 6, the total plastic-zone volume in the Multi-cavern scheme is 2,344 m^3, of which approximately 2,243 m^3 occurred in the rock mass and 101 m^3 occurred in the concrete. The plastic zones in the rock mass are distributed near the cavern walls, with almost no plastic zones observed at the cavern roof. The plastic zones in the concrete liner are mainly distributed along the Y-direction at the flanks of each cavern. The total plastic-zone volume in The Single-cavern scheme is 12,836 m^3, of which 9,571 m^3 occurred in the rock mass and 3,265 m^3 occurred in the concrete. Although plastic zones are also present along the cavern walls, the roof exhibits significantly more plastic zones than other locations. In the concrete liner, plastic zones are more pronounced at both the top and bottom of the LRC. The total plastic-zone volume in the Multi-cavern scheme is only 18 % of that in the Single-cavern scheme. Therefore, under the same maximum operating gas pressure condition, the Multi-cavern scheme demonstrates superior mechanical stability compared with Single-cavern scheme.

Fig. 7 illustrates the principal-stress distributions for the two benchmark cases, with color indicating the magnitude of the minimum principal stress. The regions surrounding the caverns are dominated by tensile stress, corresponding to the tensile failure plastic zones shown in Fig. 6. In Fig. 7a, Side View 1 represents the cross-section of the middle cavern in the Multi-cavern scheme, while Side View 2 shows the cross-section of the concrete pillar between sub-caverns. As can be seen from Fig. 7a, radially distributed tensile-stress concentrations occur in the concrete pillars between sub-caverns, indicating that these areas are prone to tensile failure. In practice, these areas should be adequately reinforced.

4.2. *Effect of cavern spacing in the Multi-cavern scheme*

In the numerical models, varying the sub-cavern spacing alters not only the distance between adjacent sub-caverns but also the distance to the two ends of the main cavern, given a fixed number of five sub-caverns. To eliminate end effects, the plastic zone volumes at both ends of the cavern were not considered in this subsection.

As shown in Fig. 8a, starting from the base case with a spacing of 10 m, the maximum tensile stress in the steel liner increases nonlinearly as cavern spacing decreases. Compared with the case of cavern spacing S_c

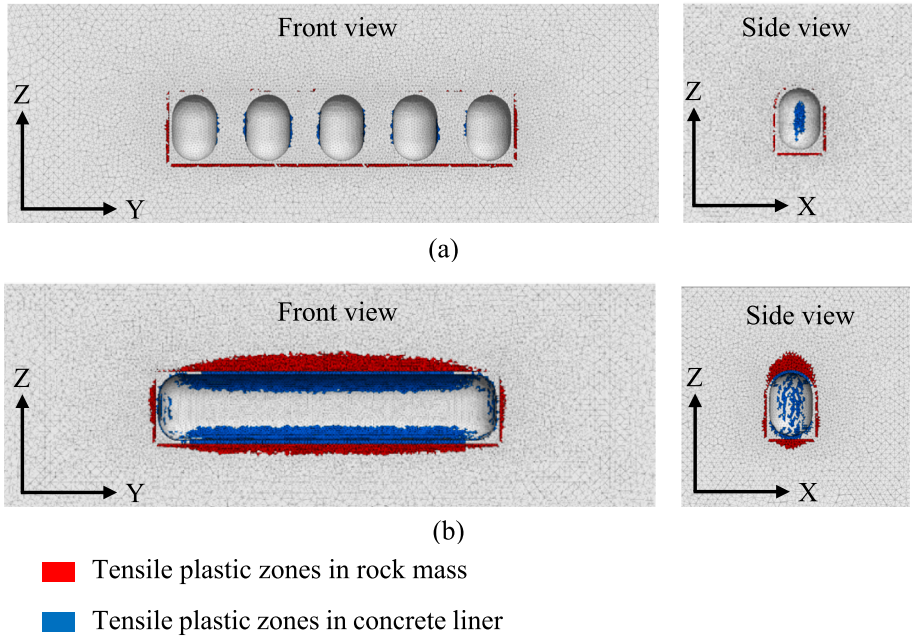


Fig. 6. Cross-section of plastic zones: (a) benchmark case of the Multi-cavern scheme; (b) benchmark case of the Single-cavern scheme.

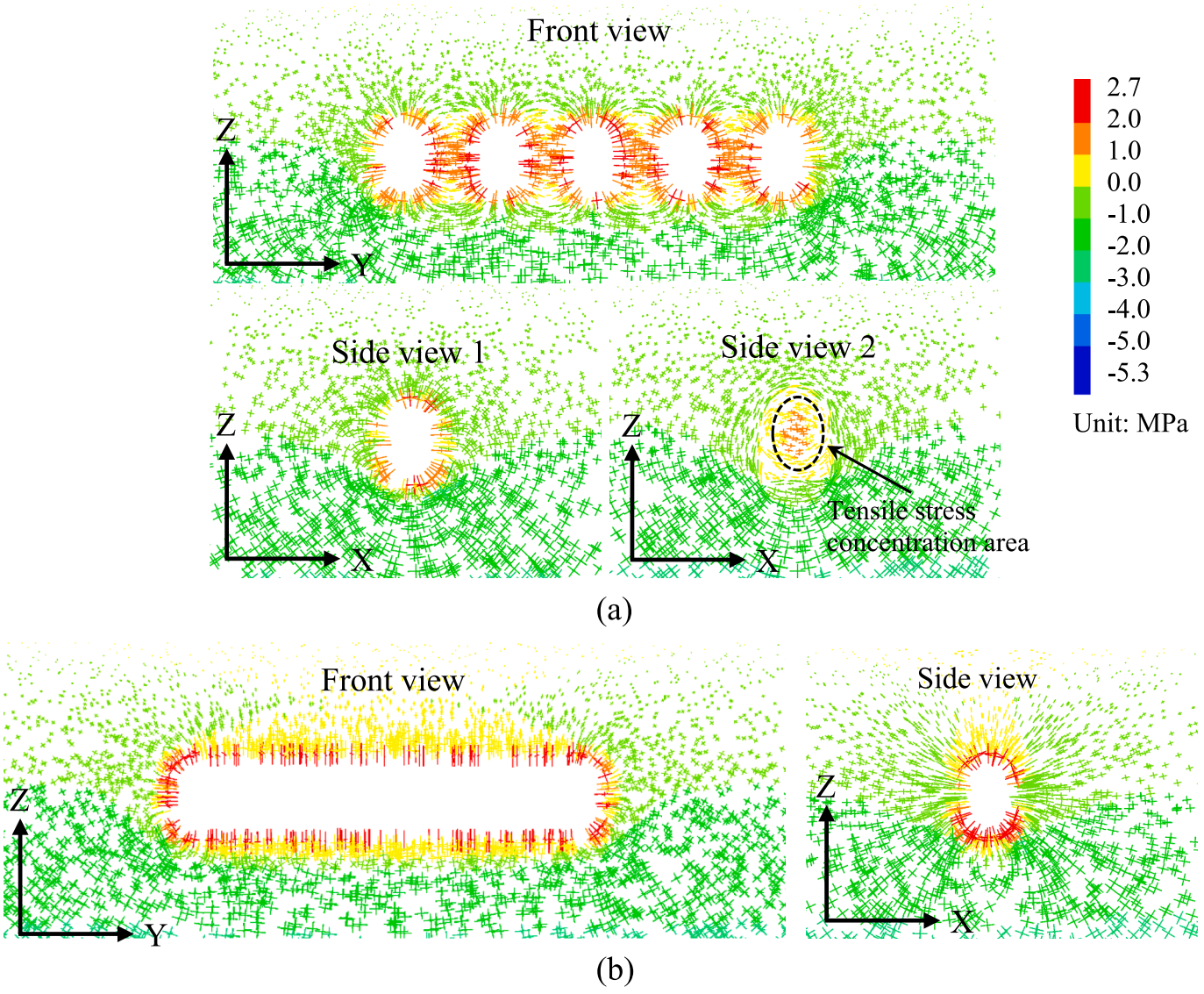


Fig. 7. Principal stress tensor at the cross-section and the magnitude of minimum principal stress: (a) benchmark case of the Multi-cavern scheme; (b) benchmark case of the Single-cavern scheme.

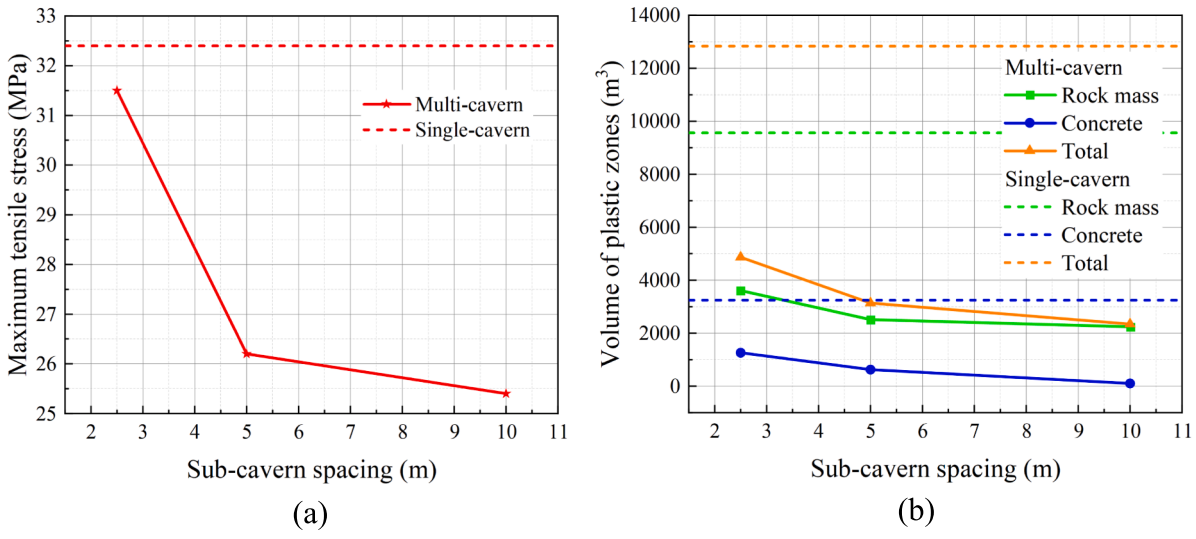


Fig. 8. Effects of sub-cavern spacing on (a) the maximum tensile stress in the steel liner and (b) plastic-zone volume.

= 10 m, the maximum tensile stress in the steel liner increases by 3 % and 24 % for $S_c = 5$ m and $S_c = 2.5$ m, respectively. Fig. 8b illustrates the variation in plastic zone volume as a function of cavern spacing. The plastic-zone volumes in both the rock mass and concrete decreased monotonically with increasing spacing. Compared with the case of cavern spacing $S_c = 10$ m, the plastic zone volume increases by 34 % and 108 % for $S_c = 5$ m and $S_c = 2.5$ m, respectively. Specifically, the plastic zone volume in the rock mass increased by 12 % and 61 %, respectively. And the corresponding increases in the concrete were 518 % and 1,135 %. In general, when $S_c = 2.5$ m, the maximum tensile stress in the steel liner in the Multi-cavern scheme is close to that in Single-cavern scheme, but the total plastic zone volume of the Multi-cavern scheme is only 38 % of that of Single-cavern scheme. This further reflects that the Multi-cavern scheme is more stable than Single-cavern scheme. Although reduced spacing between sub-caverns enhances storage layout efficiency and concrete utilization, it also intensifies interaction-induced damage in the pillars; therefore, adequate pillar reinforcement and compliance with liner stress limits are required.

4.3. Parameter sensitivity analysis

To systematically assess the influence of five key parameters on the mechanical response of the LRC structure, a parametric sensitivity analysis was conducted within reasonable engineering variation ranges, based on the context of this study. The results are shown in Fig. 9. All parameters are referenced to the benchmark case of Single-cavern scheme, and the responses are compared in a normalized form.

As shown in Fig. 9a, the maximum tensile stress in the steel liner exhibits distinctly different sensitivities to the investigated parameters. Among them, the *GSI* has the most significant effect. The effects of cavern burial depth, in-situ stress ratio, and the steel- and concrete-liner thicknesses correspond to only 22.3 %, 2.4 %, 1.6 %, and 8.3 % of the influence of *GSI*, respectively. This is primarily because *GSI* directly reflects the rock mass integrity and deformation modulus; reductions in *GSI* and associated modulus substantially weaken the confinement provided by the surrounding rock under internal pressure, thereby transferring a greater portion of the load to the steel liner. In contrast, cavern burial depth and the in-situ stress ratio have secondary effects on the stress in the steel liner. These parameters primarily alter the initial stress field of the surrounding rock and thus regulate the liner stress state in a more global manner. Variations in steel liner thickness and concrete liner thickness have relatively minor effects on the maximum tensile stress, indicating that, within the investigated thickness ranges, the steel liner stress is governed predominantly by external rock mass conditions rather than by structural dimensions.

Fig. 9b shows that the plastic-zone volume in the rock mass is highly sensitive to the cavern burial depth, *GSI*, and in-situ stress ratio. In contrast, the steel- and concrete-liner thicknesses have a relatively small impact, indicating that the steel liner and concrete liner carry only a limited portion of the internal pressure and have a limited restraining effect on the plastic-zone volume in the rock mass. Taking cavern burial depth as the reference parameter, the effects of *GSI*, in-situ stress ratio,

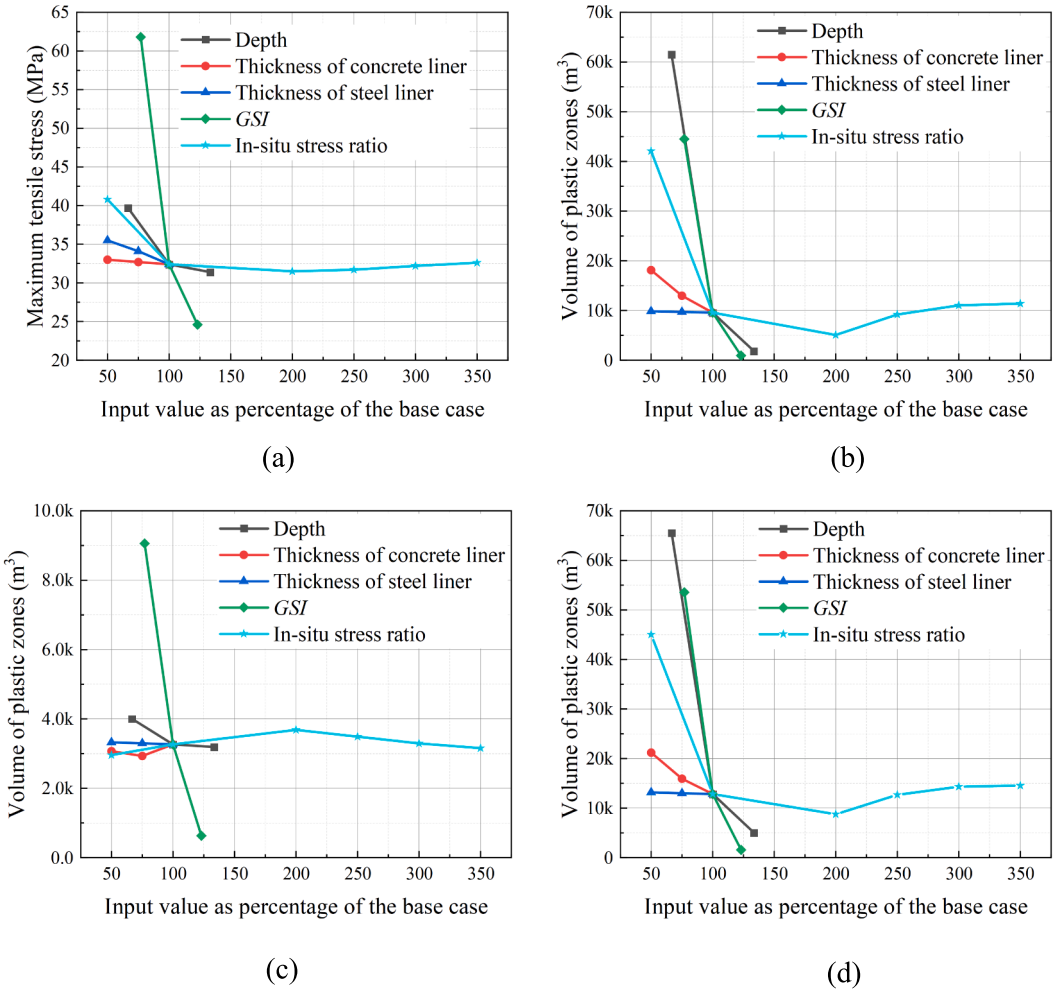


Fig. 9. Results of the parametric study: (a) maximum tensile stress of steel liner; (b) plastic-zone volume in rock mass; (c) plastic-zone volume in concrete liner; (d) total plastic-zone volume.

and the thickness of the steel and concrete liners correspond to 73.1 %, 3.7 %, 14.4 %, and 0.5 % of its influence, respectively.

As shown in Fig. 9c, the volume of plastic zones in the concrete liner is most sensitive to variations in *GSI*, while its sensitivity to burial depth, liner thickness, and in-situ stress ratio is relatively low. This suggests that the development of plasticity in the concrete liner is largely influenced by the deformation compatibility with the surrounding rock; as rock mass quality deteriorates, the liner becomes more susceptible to stress concentration and localized yielding. Using *GSI* as the reference parameter, the influences of cavern burial depth, in-situ stress ratio, and the steel- and concrete-liner thicknesses correspond to 9.6 %, 3.9 %, 3.9 %, and 0.7 % of the *GSI* influence, respectively.

Fig. 9d shows that the total plastic-zone volume is highly sensitive to the cavern burial depth, *GSI*, and in-situ stress ratio. In contrast, the thickness of the steel and concrete liners has a relatively minor impact. Taking cavern burial depth as the reference, the effects of *GSI*, in-situ stress ratio, and the thickness of the steel and concrete liners are 86.1 %, 3.1 %, 13.8 %, and 0.6 % of its influence, respectively.

Overall, the response of both the steel and concrete liners is governed primarily by *GSI*, while the bearing capacity of the rock mass is mainly controlled by *GSI* and cavern burial depth, with a lesser contribution from the in-situ stress ratio. Notably, variations in the in-situ stress ratio within the range of 1 to 3.5 have only a limited effect. Changes in liner thickness have relatively little effect on the results, suggesting potential for cost optimization. In addition, it should be noted that changing the concrete liner thickness also alters the total volume of the concrete liner, which may complicate the interpretation of using plastic zone volume as a performance indicator for the LRC. Therefore, the results presented in Fig. 9c for different concrete liner thicknesses require further evaluation. As illustrated in Fig. 10, a reduction in concrete liner thickness leads to an increase in the ratio of the plastic zone volume to the total concrete volume, indicating a higher degree of damage within the concrete liner.

5. Preliminary material cost evaluation

Gas-tightness is a key requirement for the conversion of an existing cavern into an LRC for hydrogen storage. In the present assessment, the steel liner is treated as the primary gas-tight barrier and is assumed to remain intact. Numerical results reported by Long and Ye (2025) indicate that hydrogen permeation through the intact 4130X steel liner considered in their study can remain low and can be further reduced by increasing the liner thickness. However, long-term gas-tightness depends on several factors, including steel grade, liner thickness, weld integrity, material defects, and material degradation. In particular, hydrogen diffusion through S355 steel may warrant further consideration for long-term operation (Sun et al., 2024). Accordingly, the present cost assessment includes the initial material cost of the steel liner but does not quantify the long-term costs associated with hydrogen loss, integrity monitoring, maintenance, or liner repair and replacement. The economic results should therefore be interpreted as a preliminary material-cost assessment under the assumption of an intact steel liner.

Based on the results of the numerical parametric sensitivity analyses, *GSI*, cavern burial depth *H*, and in-situ stress ratio are identified as the dominant factors influencing the plastic-zone volume. However, the effect of the stress ratio becomes less significant when it exceeds unity, as is typical for shallow excavations in Scandinavia. Therefore, the subsequent preliminary cost analysis focuses primarily on the influence of *GSI* and *H*. In addition, as the adopted uplift failure model explicitly accounts for cavern length, this parameter is also incorporated into the analysis.

This section presents the methodology for the preliminary cost evaluation, considering only the primary material components, namely the concrete liner and steel liner. Construction costs are not included because they depend strongly on project-specific conditions and are affected by various factors such as construction methods, contractor organization, site conditions, construction scheduling, and local market conditions. The quantities of these materials were determined based on the geometric characteristics of the cavern. The analysis was conducted using the benchmark cases of two schemes. Similarly, a $F_{sf} = 3$ was adopted, and the maximum operating gas pressure was calculated using Eq. (3), from which the corresponding hydrogen storage mass is obtained. Finally, a unit material-cost metric (EUR/kg H_2) is derived to facilitate comparative evaluation.

5.1. Material cost evaluation methodology

To provide a preliminary evaluation of the differences in material efficiency between the two schemes under the same gas storage pressure, a cost calculation method was developed to estimate the steel and concrete usage per unit mass of stored hydrogen. The method is based on three key factors: storage pressure, steel liner surface area, and concrete volume, while construction costs are neglected. The gas storage pressure can be calculated using parameters from the benchmark case of the Single-cavern scheme, with a $F_{sf} = 3$, and the steel liner surface area and concrete volume can be calculated based on the geometry of both benchmark cases. The mass of hydrogen that can be stored in a cavern is

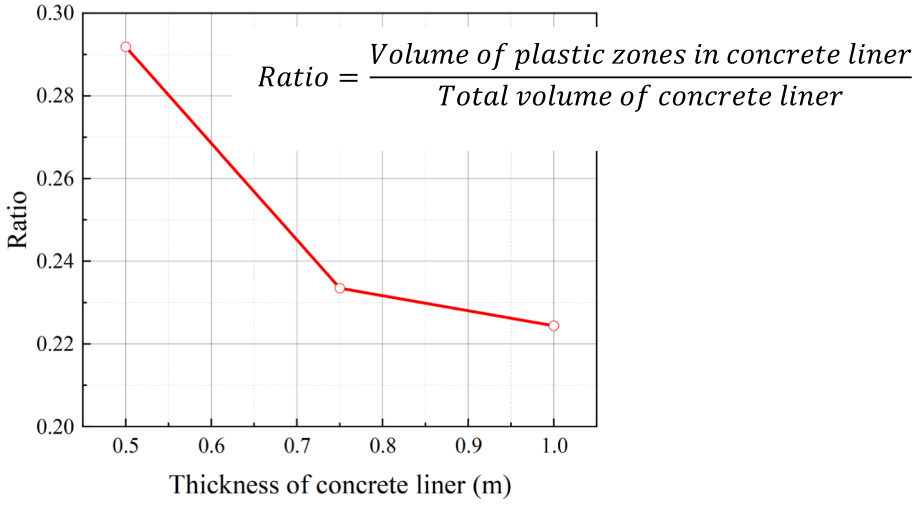


Fig. 10. The ratio of plastic-zone volume to total concrete volume versus the concrete liner thickness.

given by:

$$m_H = \frac{PV \times 10^6}{ZRT} \tag{5}$$

where P denotes the design gas storage pressure of the cavern, MPa; V is the effective gas storage volume of the cavern, m^3; Z is the compressibility factor of hydrogen and varies with temperature and gas pressure (Hu et al., 2026), but cyclic temperature variation was not considered in this calculation; R is the specific gas constant for hydrogen, 4.124×10^3 J/(kg·K); and T is the storage temperature, assumed to be the ambient temperature of 293.15 K.

Z is given by Hu et al. (2024):

$$Z = 1 - \frac{9P}{128P_H}\frac{T_H}{T}\left(\frac{6T_H^2}{T^2} - 1\right) \tag{6}$$

where P_H is the critical pressure of hydrogen, equal to 1.3 MPa; T_H is the critical temperature of hydrogen, equal to 33 K (Ibrahimoglu et al., 2021).

Therefore, the total material cost per unit mass of stored hydrogen is equal to the concrete liner and steel liner costs, each expressed per unit mass of hydrogen:

$$C_t = C_c + C_s = \frac{V_c U_c}{m_H} + \frac{A_s U_s}{m_H} \tag{7}$$

where C_t represents the total material cost per unit mass of stored hydrogen, EUR/kg H_2; C_c and C_s denote the concrete liner and steel liner costs per unit mass of stored hydrogen, respectively; V_c is the volume of concrete, m^3; A_s is the surface area of the steel liner, m^2; U_c denotes the price per unit volume of concrete, EUR/m^3; U_s denotes the price per unit area of steel liner, EUR/m^2.

Combining Eqs. (5) and (7) shows that C_t is inversely proportional to the gas pressure.

$$C_t = \frac{ZRT}{PV \times 10^6}(V_c U_c + A_s U_s) \tag{8}$$

5.2. Material cost analysis

Based on the design parameters of the cavern in the benchmark cases of the Multi-cavern and Single-cavern schemes, the Multi-cavern scheme requires 32,898 m^3 of concrete and 5,781 m^2 of steel liner, while providing an effective storage volume of 17,761 m^3. The Single-cavern scheme requires 14,548 m^3 of concrete and 7,840 m^2 of steel liner, with an effective storage volume of 36,110 m^3. For an indicative relative comparison, the historical unit prices reported by Kravanja and Žlender (2012) were adopted without adjustment to current price levels: U_c = 190 EUR/m^3 and U_s = 920 EUR/m^2, respectively; the steel price corresponds to a 12 mm thick S355 liner. Using P = 3.3 MPa together with Eqs. (5) and (6), the masses of hydrogen that can be stored in the Multi-cavern and Single-cavern schemes are 48,480 kg and 98,569 kg, respectively. Using P = 3.3 MPa and Eqs. (6) and (8), the corresponding material costs per unit mass of hydrogen are calculated as 243 EUR/kg H_2 for the Multi-cavern scheme and 103 EUR/kg H_2 for the Single-cavern scheme. The resulting unit material cost of the Single-cavern scheme is significantly lower than that of the Multi-cavern scheme. The use of concrete pillars between sub-caverns in the Multi-cavern scheme substantially increases concrete consumption, which is a key contributor to the higher cost.

For the benchmark case of the Single-cavern scheme, Fig. 11a and b illustrate the variation in gas pressure P, hydrogen storage mass m_H, and unit material cost C_t with respect to GSI and cavern burial depth H, respectively. Both P and m_H increase monotonically with increasing GSI and H. This trend arises because higher GSI and greater H permit higher maximum operating gas pressures according to the uplift failure model, thereby increasing the hydrogen storage capacity. In contrast, the unit

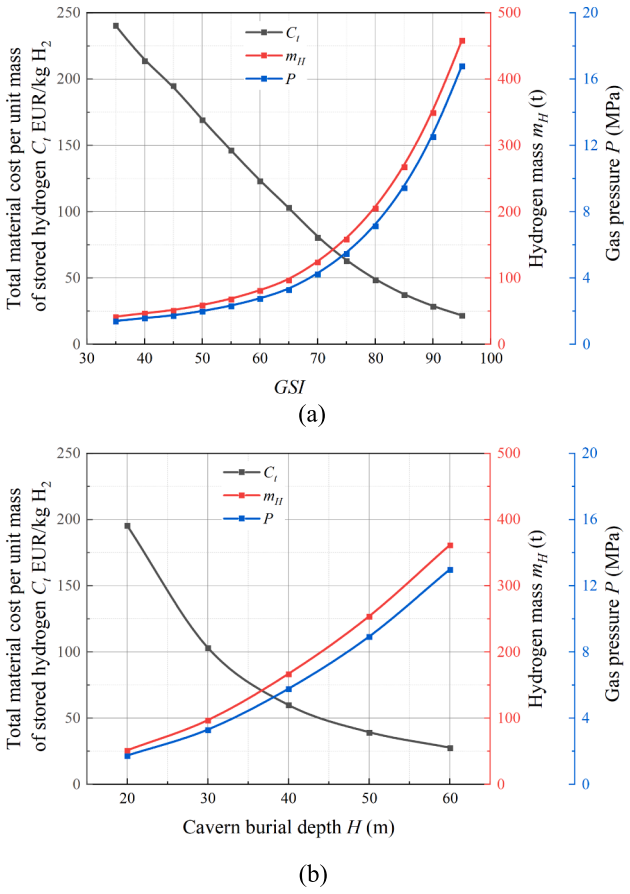


Fig. 11. Variation of total material cost per unit mass of stored hydrogen with (a) GSI and (b) cavern burial depth H.

material cost C_t exhibits a pronounced decreasing trend with increasing GSI and H. This behavior is primarily attributed to the fact that, for a given cavern geometry, the quantities of construction materials for the LRC (i.e., concrete and steel liner) remain nearly constant, while the storage capacity increases significantly with gas pressure. Consequently, the cost per unit mass of stored hydrogen decreases. This observation is consistent with Eq. (8), which indicates an inverse relationship between C_t and P.

Further analysis using the design of the Single-cavern scheme and F_{sf} = 3 allows the assessment of other factors influencing the material cost per unit mass of stored hydrogen. Fig. 12 shows that the effect of cavern length on the total material cost per unit mass of stored hydrogen increases gradually. As the cavern length increases from 122 m to 322 m, the cavern volume increases, whereas the maximum operating gas pressure (as defined by Eq. (3)) decreases. Since material cost scales with cavern volume, while the total stored hydrogen mass is governed by both the increase in volume and the reduction in pressure, the rate of cost increase exceeds that of the stored mass. Consequently, the material cost per unit mass of stored hydrogen increases with increasing cavern length.

Overall, maximum operating gas pressure plays a dominant role in determining the total material cost per unit mass of stored hydrogen. Greater burial depth allows higher gas pressure and therefore reduces the unit material cost. Increasing cavern length raises total storage capacity but reduces the maximum operating pressure and slightly increases the unit material cost. For the Multi-cavern scheme, a reasonable sub-cavern spacing must be set to control costs. Consequently, the design of repurposed oil caverns for underground hydrogen storage should prioritize maximizing operating gas pressure within safety and regulatory limits, while efficiently utilizing burial depth. In addition,

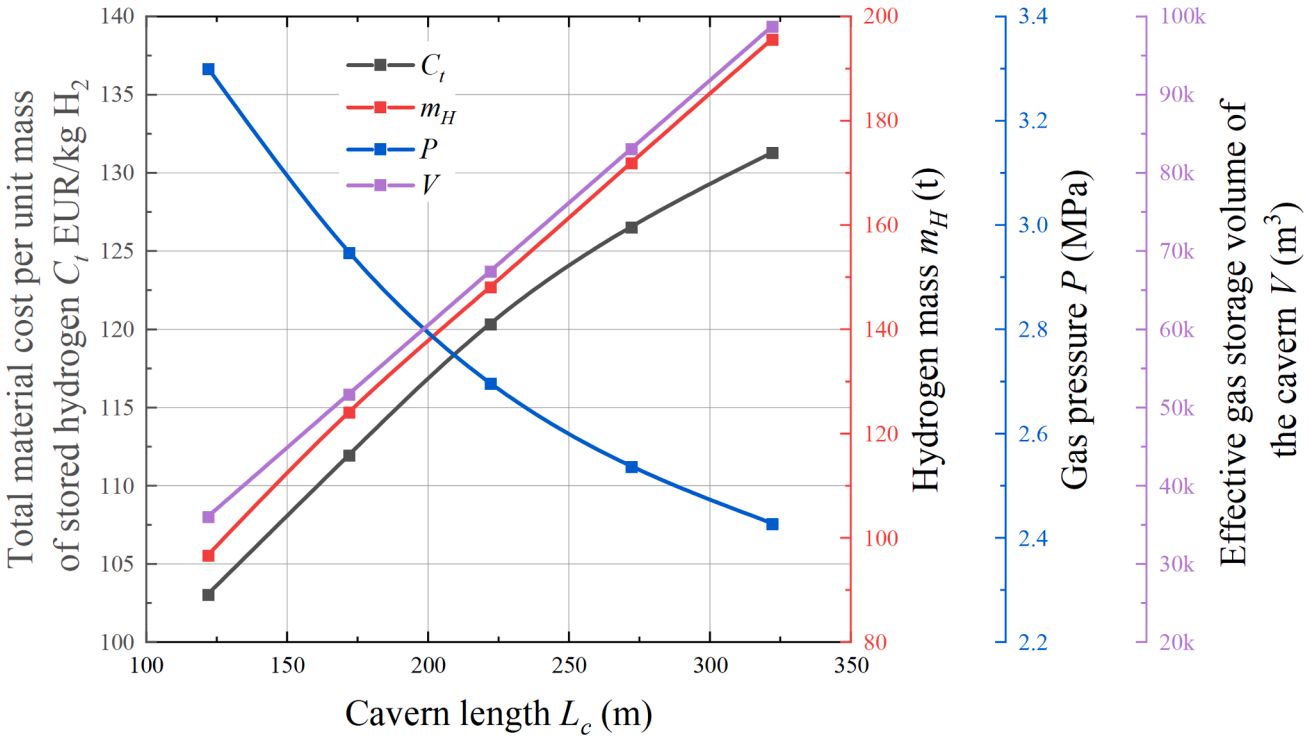


Fig. 12. Effect of cavern length on total material cost per unit mass of stored hydrogen.

shorter caverns are preferable when minimizing C_t, whereas longer caverns are advantageous when maximizing total hydrogen storage capacity, despite their reduced maximum operating gas pressure and higher unit cost.

6. Discussion

The present study evaluated the feasibility of repurposing oil storage caverns for hydrogen storage using LRC technology, with primary emphasis on the mechanical interaction among the surrounding rock mass, concrete liner, and steel liner. Building on this assessment, the Discussion further examines the short-term thermo-mechanical response of the LRC system under cyclic gas-pressure and gas-temperature loading during representative hydrogen injection–withdrawal cycles. It also addresses the engineering applicability and limitations of the adopted methodology.

6.1. Evaluation of thermo-mechanical response under cyclic operation

Although the primary objective of this study is to evaluate the mechanical and structural feasibility of retrofitting oil storage caverns into lined rock caverns (LRCs) under static loading conditions, hydrogen storage caverns in practice operate under repeated injection and withdrawal cycles. These cycles induce simultaneous variations in cavern gas pressure and temperature, which may influence the thermo-mechanical response of the steel liner and surrounding materials. To examine whether the static analyses presented in Sections 3 and 4 provide a reasonable representation of operational conditions, a supplementary thermo-mechanical analysis was conducted.

A thermodynamic model based on mass and energy conservation was used to calculate the transient gas pressure and temperature histories during cyclic hydrogen injection and withdrawal. The governing equations presented in Eqs. (9)–(11) were based on the formulations reported by Kushnir et al. (2012) and Zhou et al. (2015, 2017). The cavern geometry, operating conditions, material properties, and thermodynamic parameters were specified for the present simulations. The calculated gas-pressure and gas-temperature histories were subsequently applied as prescribed boundary conditions on the cavern wall in FLAC3D to evaluate the structural response of the steel liner (Jiang et al., 2020). This procedure represents a one-way sequential thermo-mechanical analysis, in which the thermodynamic response governs the mechanical loading, whereas the influence of structural deformation on the gas thermodynamics is neglected.

$$\frac{V\mathrm{d}\rho}{\mathrm{d}t} = (F_i + F_e)\dot{m}_c \tag{9}$$

$$V\rho c_{v0}\frac{\mathrm{d}T}{\mathrm{d}t} = F_i\dot{m}_c\left[c_{p0}(T_i - T) + Z_0RT + \rho\frac{RT_0^2Z_{T0}}{\rho_0}\right] + F_e\dot{m}_c\left(Z_0RT + \rho\frac{RT_0^2Z_{T0}}{\rho_0}\right) + A_ch_c(T_{r0} - T) \tag{10}$$

$$\begin{cases} F_i = 1 \text{ and } F_e = 0, & \text{during charging stage} \\ F_i = 0 \text{ and } F_e = -2, & \text{during dischargingstage} \\ F_i = 0 \text{ and } F_e = 0, & \text{during storage stage} \end{cases} \tag{11}$$

where ρ and ρ_0 are the density and initial density of hydrogen in the cavern, respectively; $\dot{m}_c$ is the mass flow rate of hydrogen (kg/s); R is the gas constant for hydrogen; the subscripts i and e denote the gas state in the inlet and outlet of the cavern, respectively; T_{r0} is the temperature at cavern wall surface; and T_i represents the temperature of hydrogen entering the cavern at the charging stage; c_{v0}, c_{p0}, Z_0 and Z_{T0} are the values of c_v, c_p, Z and $\partial Z/\partial T$ calculated in the initial state (T_0, ρ_0). Here, c_v and c_p denote the specific heat at constant volume and specific heat at constant pressure, respectively. A_c is the surface area of the cavern wall ($A_c = A_s$); h_c is the heat exchange coefficient between the hydrogen in the cavern and the cavern wall.

Before applying the procedure to the proposed retrofit schemes, the implementation was verified by using the benchmark problem presented by Zhou et al. (2015). Fig. 13a shows the verification model constructed

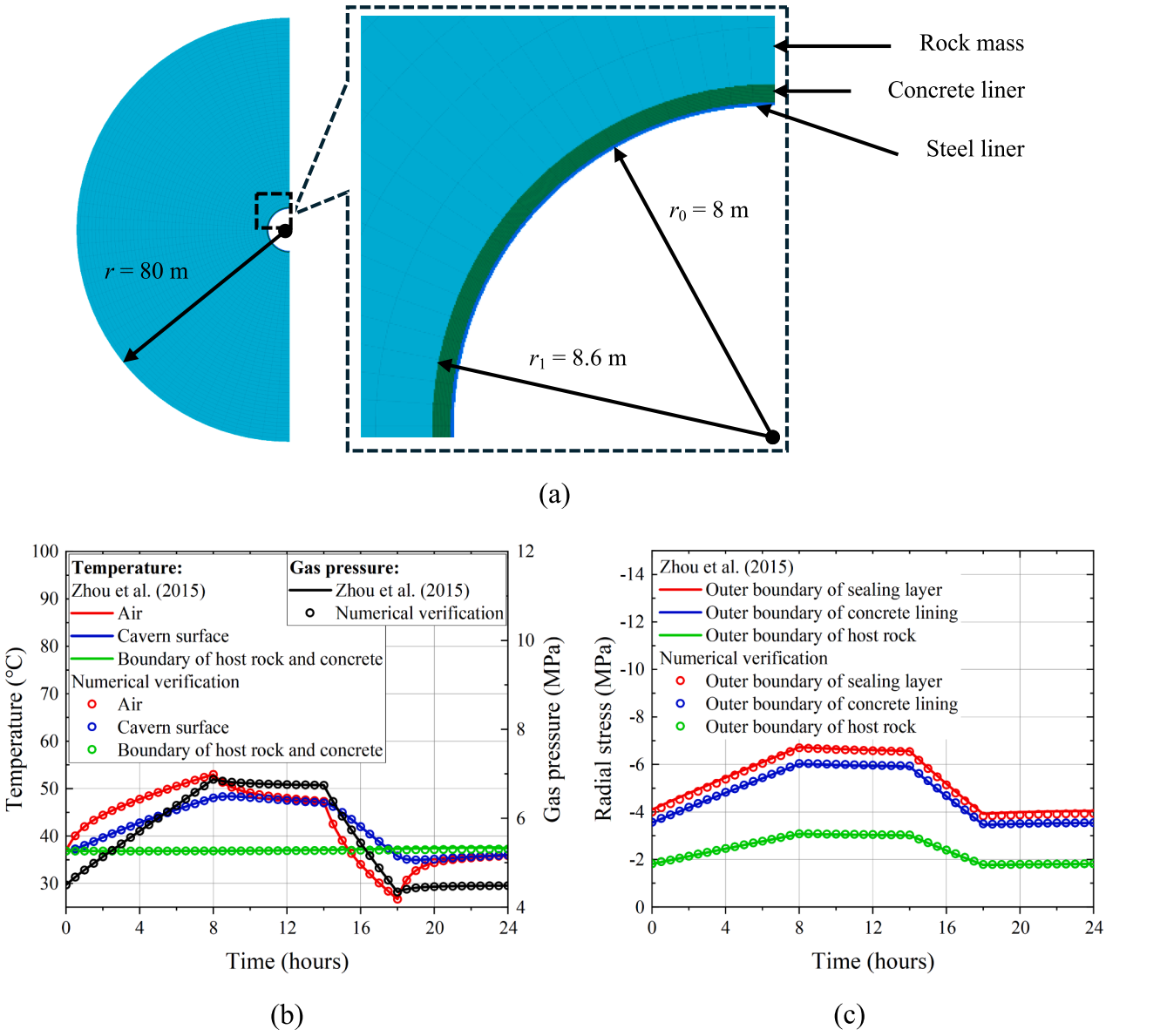


Fig. 13. Verification of the one-way sequential thermo-mechanical analysis: (a) FLAC3D model; (b) comparison of calculated gas-pressure and gas-temperature histories; and (c) comparison of radial stresses obtained from the analytical and numerical solutions.

in FLAC3D. The predicted gas-pressure and gas-temperature histories agree closely with the published solutions (Fig. 13b), while the calculated radial stresses show good agreement with the corresponding analytical solution (Fig. 13c). These comparisons verify the implementation of the thermodynamic model, the transfer of the transient pressure and temperature histories to the numerical model, and the resulting thermo-mechanical response.

Following the benchmark verification, the same procedure was applied to the long Single-cavern model shown in Fig. 4b. Damasceno (2022) reported a representative two-week hydrogen injection–withdrawal cycle for LRC operation, which was adopted in this study. The relevant thermophysical properties of the steel liner, concrete liner, rock mass and hydrogen gas are summarized in Table 5 and Table 6. Two representative operating conditions, with maximum gas pressures P_{max} of 3.3 and 7.0 MPa, were analyzed. These correspond to the benchmark operating pressure and the highest operating pressure obtained in the parametric study cases presented in Section 3.3. Each simulation covered two consecutive hydrogen injection–withdrawal cycles, with each cycle lasting two weeks. The initial gas pressure, temperature field, and in-situ stress field were specified consistently with the benchmark model, assuming a cavern depth of 30 m. The initial temperature profile was defined using a constant surface temperature of 10.0 ℃ and a vertical temperature gradient of 0.03 ℃/m, giving an initial temperature of approximately 11.0 ℃ at the cavern depth.

Table 5
Thermophysical parameters.

	Thermal conductivity W/(m•K)	Specific heat J/(kg·K)	Coefficient of thermal expansion 1/K	Heat transfer coefficient W/(m^2·K)
Steel liner	45	500	1.7×10^{-5}	50
Concrete liner	1.4	837	1.2×10^{-5}	–
Rock mass	3.5	1000	1.2×10^{-5}	–

Fig. 14 presents the calculated histories of cavern gas pressure, gas temperature and the maximum tensile stress in the steel liner. For both operating pressures, the maximum tensile stress occurs when the cavern

Table 6
Hydrogen gas parameters.

Parameters	Symbol	Value	Unit
Initial pressure in the cavern	P_0	1.0	MPa
Maximum pressure	P_{max}	3.3, 7.0	MPa
Initial temperature in the cavern	T_0	11.0	℃
Injected temperature at cavern port	T_i	15.0	℃
Mass flow rate through the compressor	$\dot{m}_c$	0.167, 0.420	kg/s
Initial value of specific heat at constant pressure	c_{p0}	14,304	J/(kg·K)
Initial value of specific heat at constant volume	c_{v0}	10,180	J/(kg·K)

Note: P_{max} = 3.3 MPa and 7.0 MPa correspond to $\dot{m}_c$ = 0.167 kg/s and 0.420 kg/s, respectively.

gas pressure reaches its minimum, consistent with the observation reported by Zhou et al. (2015). More importantly, the peak tensile stress obtained under cyclic thermo-mechanical loading is considerably lower than that predicted by the corresponding static analysis, in which the maximum operating pressure was applied as a constant internal load.

Fig. 15 compares the plastic-zone volumes obtained from the static and cyclic thermo-mechanical analyses. At P_{max} = 3.3 MPa, cyclic pressure and temperature loading reduced the plastic-zone volume in the concrete liner by 63.7 % but increased that in the surrounding rock by 42.4 %, resulting in a 15.4 % increase in the total plastic-zone volume. At P_{max} = 7.0 MPa, the corresponding changes were 29.3 %, 30.3 %, and 19.6 %, respectively. These contrasting trends suggest that cyclic thermo-mechanical loading redistributes plastic yielding between the concrete liner and surrounding rock, rather than uniformly reducing or increasing the extent of plasticity.

This comparison highlights the response-dependent differences between the constant-pressure static loading adopted in the primary analyses and the short-term cyclic gas-pressure and gas-temperature loading. For the cases examined, the static analysis overestimated the maximum tensile stress in the steel liner and the concrete plastic-zone volume, while underestimating the rock-mass and total plastic-zone volumes. Therefore, the static analysis does not conservatively bound all response quantities, particularly those associated with rock-mass yielding. Nevertheless, given its simplicity and computational efficiency, the static approach remains useful for preliminary screening and comparative parametric assessment, provided that its response-dependent limitations are recognized and an adequate safety margin is maintained when determining the maximum operating gas pressure.

A detailed cyclic thermo-mechanical assessment requires specification of the operational conditions, including hydrogen injection and withdrawal rates, cycle duration, pressure range, and gas injection temperature, as these parameters govern the transient pressure and temperature histories and consequently influence the thermo-mechanical response of the cavern system. Such operational parameters are often not well defined at the initial feasibility stage, particularly when assessing the potential for repurposing existing oil storage caverns for underground hydrogen storage. The static approach adopted in the primary analyses is therefore intended as a preliminary assessment method applicable when detailed operational information is not yet available, whereas cyclic thermo-mechanical analysis becomes increasingly important as the storage design and operating strategy are further developed.

The present thermo-mechanical analysis is intended to supplement the static modeling framework rather than provide a comprehensive assessment of long-term cyclic performance. Only two representative operating cycles were considered, and the analyses were performed using a one-way sequential coupling strategy. Progressive damage accumulation, fatigue of the steel liner and concrete liner, groundwater effects, hydrogen diffusion, and the feedback of structural deformation on cavern thermodynamics were not included. Therefore, the present analyses characterize only the short-term thermo-mechanical response and cannot establish the long-term evolution of the liner–concrete–rock system. Future studies should consider extended cyclic operation and more comprehensive thermo-hydro-mechanical coupling, together with project-specific operational conditions, to evaluate progressive rock-mass degradation, liner fatigue, gas-tightness, and long-term structural reliability.

6.2. *Retrofitting schemes and materials*

The preceding analyses show that the two retrofit schemes involve a clear stability-cost trade-off. The two proposed retrofit schemes are simplified designs that primarily differ in their LRC configurations; therefore, the influence of cavern plugs was neglected. Assuming that both schemes induce identical uplift effects in the overlying rock mass, the gas pressure determined using the uplift failure model was adopted for stability comparison in the numerical simulations. However, other LRC configurations can be considered when modifying oil storage caverns. For example, a hybrid configuration using both LRC geometries from the Multi-cavern scheme and Single-cavern scheme may provide a balance between structural stability and hydrogen storage efficiency.

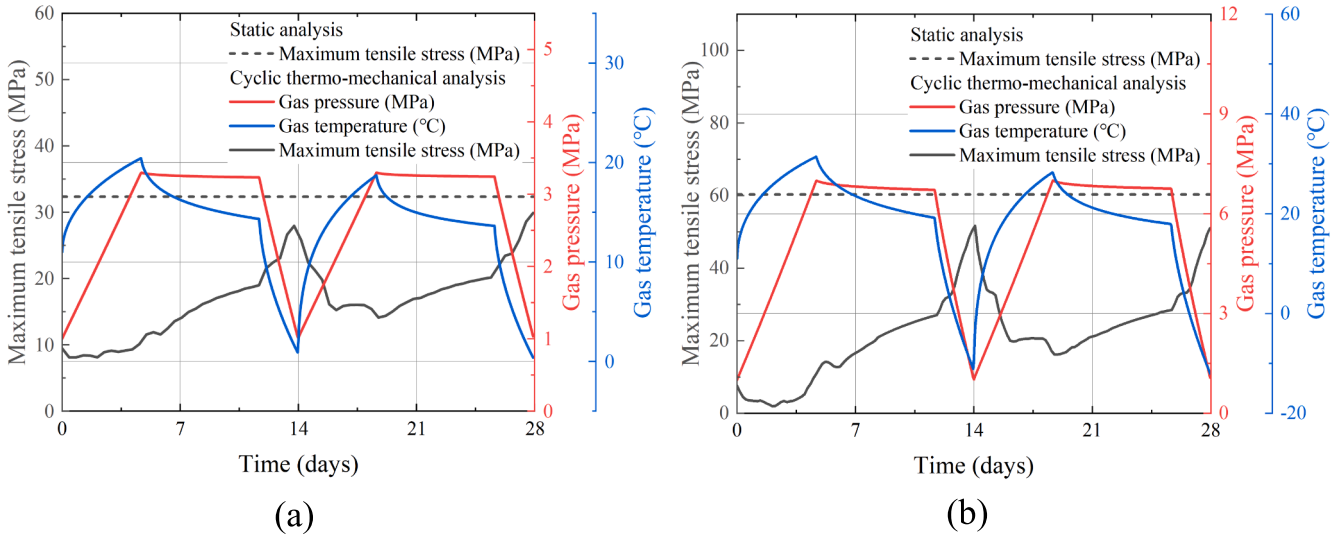


Fig. 14. Time histories of cavern gas pressure, gas temperature and the maximum tensile stress in the steel liner: (a) P_{max} = 3.3 MPa; and (b) P_{max} = 7.0 MPa.

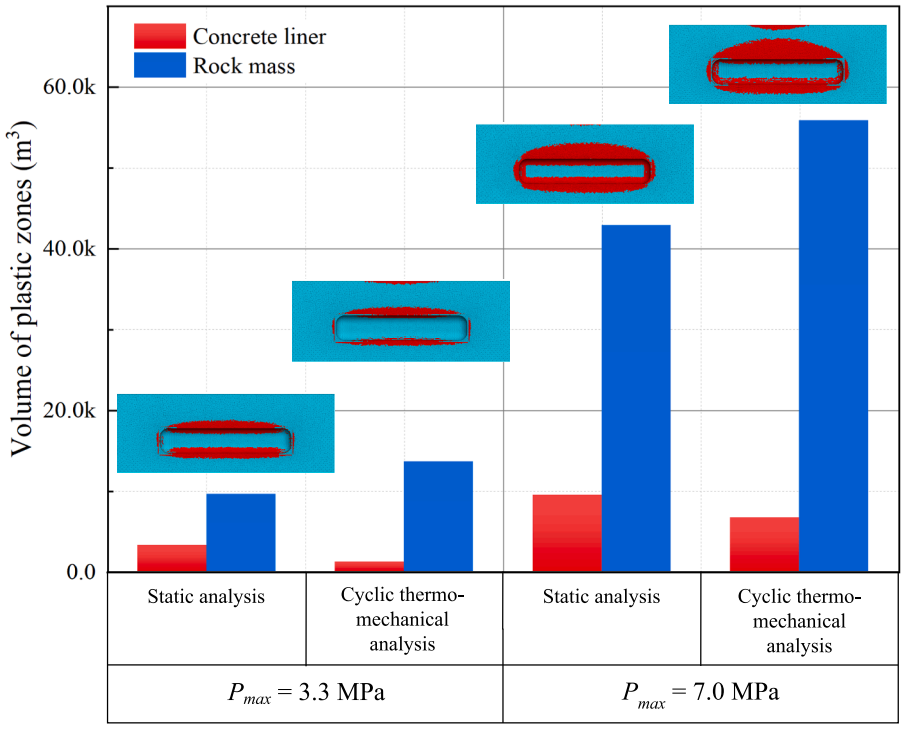


Fig. 15. Comparison of plastic-zone volume in static analyses and cyclic thermo-mechanical analyses.

This analysis assumes the use of a conventional steel liner, which aligns with existing LRC practices and offers well-defined mechanical properties, high tensile strength, and well-established gas-tightness. However, future research could explore other liner materials, particularly in hydrogen storage. Composite liners or fiber-reinforced polymer (FRP) systems may offer advantages in corrosion resistance, low weight, and potential cost reduction in installation, but their long-term performance under high-pressure hydrogen, cyclic loading, and temperature variations remains uncertain (Bhuiyan et al., 2025). Concrete-lined caverns with internal polymer membranes or sprayed sealing layers can reduce steel requirements but may face challenges in durability and repairability. Furthermore, hybrid liner systems that combine thin steel liners with load-sharing composite or prestressed concrete layers can improve structural redundancy and achieve material optimization. Moreover, Wu et al. (2026) used loose filler to replace part of the concrete as pressure-holding material, providing a cost-reduction pathway for the Multi-cavern scheme.

6.3. Applicability of the uplift failure model

To evaluate the ultimate gas pressure of shallow elongated lined rock caverns for gas storage, six uplift failure models in two-dimensional formulations are compared. Model 3, which gives intermediate pressure estimates among the six models, is selected as a comparatively conservative screening model relative to Models 4–6 and further extended to a three-dimensional formulation. Numerical simulation results indicate that the Multi-cavern scheme exhibits superior stability, suggesting it is a promising candidate for practical application. Nevertheless, further investigation into the interaction between multiple subcaverns and the overlying rock mass is essential before implementation.

The uplift failure model adopted in this study does not explicitly account for the influence of the in-situ stress ratio. The parametric analysis indicates that the effect of λ is relatively limited within the investigated range ($H = 20$–40 m, $GSI = 50$–80, and $\lambda \geq 1$), whereas the effect is relatively large when $\lambda < 1$. Therefore, for $\lambda < 1$, when using the uplift failure model under special in-situ stress conditions, a greater safety margin is required. These findings highlight the importance of further systematic investigation into the role of in-situ stress ratios in determining the ultimate gas pressure of shallow long LRCs.

In addition, when the Mohr–Coulomb parameters are derived from the Hoek–Brown criterion, the resulting cohesion and friction angle should be interpreted as equivalent parameters obtained through a linear approximation of a nonlinear strength envelope. In particular, the friction angle is often higher than typical laboratory-derived values, as it reflects the slope of the fitted envelope over a selected confining stress range rather than a true material constant. Therefore, caution is required when these equivalent Mohr–Coulomb parameters are further applied in uplift failure models for estimating the ultimate gas pressure of lined rock caverns. Since the uplift failure mechanism is sensitive to shear strength parameters, the inherent approximation in the Hoek–Brown to Mohr–Coulomb conversion may introduce additional uncertainty in the predicted ultimate gas pressure.

6.4. Limitations and future improvement of the material-cost assessment

The economic comparison presented in this study focuses on the initial material consumption of concrete and steel per unit mass of stored hydrogen, thus providing a preliminary assessment of the material efficiency between retrofit options. While this approach highlights fundamental differences in material requirements, it does not encompass the full economic complexity of retrofit hydrogen storage projects.

Several important cost components were not considered, such as excavation retrofitting, foundation reinforcement, construction logistics in confined port environments, liner installation, quality control, and commissioning. Furthermore, operating and life-cycle costs (including hydrogen losses, inspection, maintenance, liner replacement, monitoring systems, and potential downtime) can significantly impact overall economic feasibility. The impact of hydrogen storage pressure cycling (potentially affecting fatigue life and maintenance frequency) was also not taken into account.

In addition, potential variations in material properties and specifications, including concrete grade, steel liner type and thickness, as well

as fluctuations in material prices, may introduce uncertainty into the cost analysis. Moreover, the comparison results between the two schemes may differ significantly if alternative filling materials and polymer-based sealing materials are considered in place of the steel liner.

Therefore, future economic assessments should employ a full life-cycle cost analysis framework, integrating factors such as capital expenditures, operating expenses, risk-related costs, and end-of-life disposal. Incorporating uncertainty analysis and sensitivity studies will help identify key cost drivers and robust design options, thereby further improving decision-making. Combining techno-economic analysis with structural performance indicators would allow for a more comprehensive evaluation of large-scale hydrogen storage retrofit strategies.

7. Conclusion

This study investigated the feasibility of retrofitting shallow elongated underground oil storage caverns for hydrogen storage using LRC technology. The main conclusions are as follows:

(1) The predicted ultimate gas pressure increases monotonically with cavern burial depth and rock mass quality across all models, although the sensitivity to these two factors varies among models. Among the six uplift failure models examined, Model 3 gives intermediate estimates while accounting for both overburden weight and rock-mass tensile strength. With an explicit safety factor and the stated applicability limits, it provides a preliminary screening basis for shallow LRC design.

(2) In both benchmark cases, the Multi-cavern scheme exhibits greater stability than the long Single-cavern scheme, as indicated by lower maximum tensile stress in the steel liner and substantially smaller plastic zones. The plastic zone volume in the Multi-cavern scheme is only 18 % of that in the Single-cavern scheme. However, its effective storage volume is reduced to 49 % of the Single-cavern scheme.

(3) For the Multi-cavern scheme, decreasing the sub-cavern spacing significantly increases both the maximum tensile stress in the steel liner and the volume of plastic zones in the concrete pillar. The results indicate that a sub-cavern spacing of 5 m could serve as a preliminary design parameter, subject to case-specific verification and reinforcement design. These results highlight the importance of maintaining appropriate sub-cavern spacing to control stress concentration and mitigate the risk of liner failure.

(4) The maximum tensile stress in the steel liner and the volume of plastic zones in the concrete liner are primarily governed by *GSI*, whereas the volume of the rock plastic zone is strongly influenced by cavern burial depth, *GSI*, and the in-situ stress ratio. The total plastic zone volume is dominated by the rock response. Using cavern burial depth as a reference, the relative influences of *GSI*, in-situ stress ratio, and the thickness of the steel and concrete liners on the total plastic zone volume are 86.1 %, 3.1 %, 13.8 %, and 0.6 %, respectively. This indicates that the overall nonlinear deformation behavior of the LRC is mainly controlled by rock mass quality and cavern burial depth, while in-situ stress state and liner parameters play a secondary and localized role.

(5) The unit material cost exhibits an inverse relationship with gas pressure, decreasing sharply at low pressures and approaching a plateau at higher pressures. In the benchmark cases, the C_t of the Single-cavern scheme (103 EUR/kg H_2) is significantly lower than that of the Multi-cavern scheme (243 EUR/kg H_2). Further analysis indicates that the hydrogen storage capacity (m_H) is the dominant cost driver, explaining the influence of *GSI*, cavern burial depth and cavern length. Two potential material-cost optimization measures are reducing the steel liner thickness from 12 mm to 6 mm and decreasing the sub-cavern spacing in the Multi-cavern scheme. Both measures require further verification of gas-tightness, cyclic performance, and local stress concentrations.

(6) For the two short-term cyclic cases examined, the static analysis overestimated the maximum steel-liner tensile stress and concrete plastic-zone volume but underestimated the rock-mass and total plastic-zone volumes. Therefore, the static approach does not conservatively bound all response quantities. Nevertheless, it remains useful for preliminary screening and comparative parametric assessment when detailed operational conditions are not yet available, provided that its limitations are recognized and an adequate safety margin is maintained when determining the maximum operating gas pressure.

Overall, the results support further consideration of both retrofit schemes, subject to their different stability, storage-capacity, and material-cost characteristics. Project-specific geological investigation, operational definition, and long-term structural verification are required before practical implementation.

CRediT authorship contribution statement

Wenjun Luo: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ping Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Charlie Chunlin Li:** Writing – review & editing, Investigation, Conceptualization. **Yang Zou:** Visualization, Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was conducted within the project “Hydrogen, ammonia, and methanol in hydrogen hubs in the Nordic region (H2AMN)” (Project No. 2315912-0611), funded by the Nordic Council of Ministers through Nordic Energy Research (NER). Additional financial support was provided through a seed project of the Centre for Hydrogen Energy Systems Sweden (CH2ESS) at Luleå University of Technology, Sweden. The authors thank Dr. Zongze Li for insightful discussions and valuable suggestions.

Data availability

Data will be made available on request.

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